SZÉCHENYI EGYETEM
UNIVERSITY OF GYŐR

INFORMATIKA TANSZÉK
DEPARTMENT OF COMPUTER SCIENCE

Avantvia küldetésirányító rendszerének adatbázisa

Féléves feladat Adatbázis-kezelés 2. tárgyból

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1. Bevezetés

1.1. AI használat nyilatkozat

Az adatbázis adatokkal való feltöltésére a Google `gemini-3.1-pro-preview` modelljét használtam agent módban a `gemini-cli`-n keresztül, mely a gyakorlatok után többször is feltett kérdésekre adott válaszok alapján erre a célra legálisnak és jóváhagyottnak tűnt[1]. Az AI modell feladata annak a scriptnek az elkészítése volt, amely az adatokat feltölti az adatbázisba. A script a 2. számú mellékletben található (13.2).

Feladatát részben önellenőrző módon, de szükséges vezetéssel és emberi ellenőrzés mellett látta el. Iránymutatásként kizárólag saját magam által létrehozott anyagokat kapott: az előző féléves Adatbázis-kezelés 1 tárgy féléves feladatának keretében létrehozott adatbázis dokumentációját, sémáját, az Adatbázis-kezelés 2 féléves feladatának keretében előre megírt és egy korábbi, sokkal kisebb adathalmazon validált lekérdezéseket, és egy több oldalas magyarázatot kapott arról, hogy mit és hogyan várok/vár el az adatbázis az adatokkal kapcsolatban. Ezen kívül a saját betanítási adathalmazát használta a feladatának elvégzésére. Az eredményeket a lekérdezésekkel és manuálisan is ellenőriztem, és korrigáltattam, ahol szükséges volt.

Az AI alkalmazásának két oka volt:

- az adatok létrehozásának és az adatbázisba való feltöltésének nagymértékű felgyorsítása és az adathalmaz megnövelése, több napi munkát pár órásra csökkentve, és
- az AI modell képességeinek felmérése arra vonatkozóan, hogy hogyan segíthet megfelelő idő híján egy hallgatónak, a féléves feladat lényegi részének megoldása nélkül.

Az AI által generált adathalmaz és script alapvetően nem befolyásolja a féléves feladat lényegi részét és megoldási módját, vagyis a lekérdezéseket és azok használatát. Továbbá, nem hajtott végre maradandó módosítást az adatbázis feltöltése során, csupán a múltbeli dátumok bevitele miatt hajtott végre szükséges, de ideiglenes módosítást.

1.2. Feladat ismertetés

Az Avantvia, a SpaceX mintájára megálmodott, fiktív űrvállalat, melynek küldetése a világegyetem felfedezésének és az emberiség űrbeli terjeszkedésének elősegítése. A cég neve latin eredetű mozaikszó („avantia”, „via”), jelentése „előre vezető út”. Az ötletadó mintájára részben vagy egészben újrahasznosítható rakétákból álló flottát üzemeltet. A rakéták neve a szintén latin „excelsior”, jelentése „magasabbra”. A vállalat emberes küldetéseket is hajt végre, amikhez saját gyártású űrkapszulát használnak, melynek neve „viator”, vagyis „utazó”.

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[1] A gyakorlatvezető által adott válaszokból az is kiderült, hogy még nem tudott a szabályzatról. Továbbá, szeretném megjegyezni, hogy a szabályzatok ~9 órával a kész feladat első leadása után, és az AI használatának befejezése után ~12 órával érkezett meg a Neptunban.

A rakétaflotta a következő típusokból áll:

- Excelsior 1: részben újrafelhasználható, 2 fokozatú, könnyű-közepes osztályú rakéta (konfigurációtól függően)
- Excelsior 2: részben újrahasználható, 2+1 fokozatú, közepes osztályú rakéta
- Excelsior Heavy: részben újrahasználható, 2 fokozatú, nehéz osztályú rakéta, 3 db Excelsior 2 első fokozatból és 1 db Excelsior 2 második fokozatból áll.
- Excelsior Nova: teljesen újrafelhasználható, szupernehéz osztályú rakéta.

A feladat során elkészített adatbázis a küldetésirányító és információs rendszer adatbázisa, mely a küldetéseket, rakétaflottát, rakományokat, indítási és leszállási helyeket és infrastruktúrát, űrhajósokat és ezek tulajdonságait, állapotát kezeli és tartja nyilván. Képes kezelni az egyetlen hasznos terhet szállító küldetéseket, a komplex „rideshare” repüléseket (amik sok különböző hasznos terhet szállítanak akár különböző pályákra), és képes kezelni az emberi űrrepülések teljes legénységi nyilvántartását is, szerepeket rendelve minden fedélzeti és tartalék űrhajósnak.

Az adatbázis része lesz egy jövőbeli, nagyobb személyes projektnek, amely a teljes rakétaipar küldetéseit képes lesz hasonló módon kezelni, ezért ez a projekt annak alapjaként lett kialakítva. A dokumentumban, ahol szükséges, ezzel kapcsolatos megjegyzések kerülnek beszúrásra.

1.3. Funkcionális követelmények

Az adatbázis a következő funkcionális követelményeknek kell, hogy megfeleljen:

- Adatok felvitele a felhasználó által
- Adatok módosítása a felhasználó által
- Adatok törlése a felhasználó által
- Küldetések állapotának megjelenítése

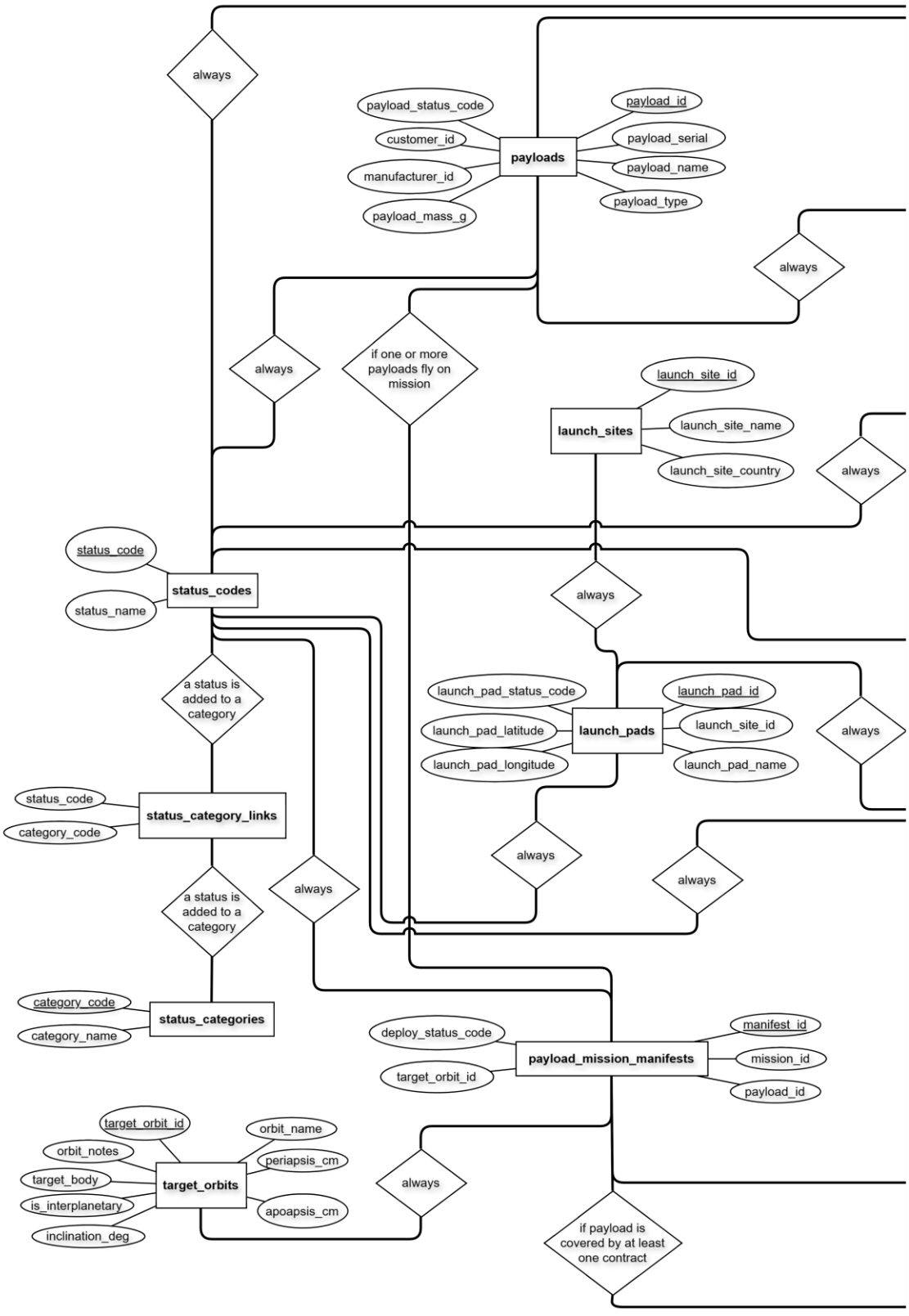
1.4. Adathalmaz

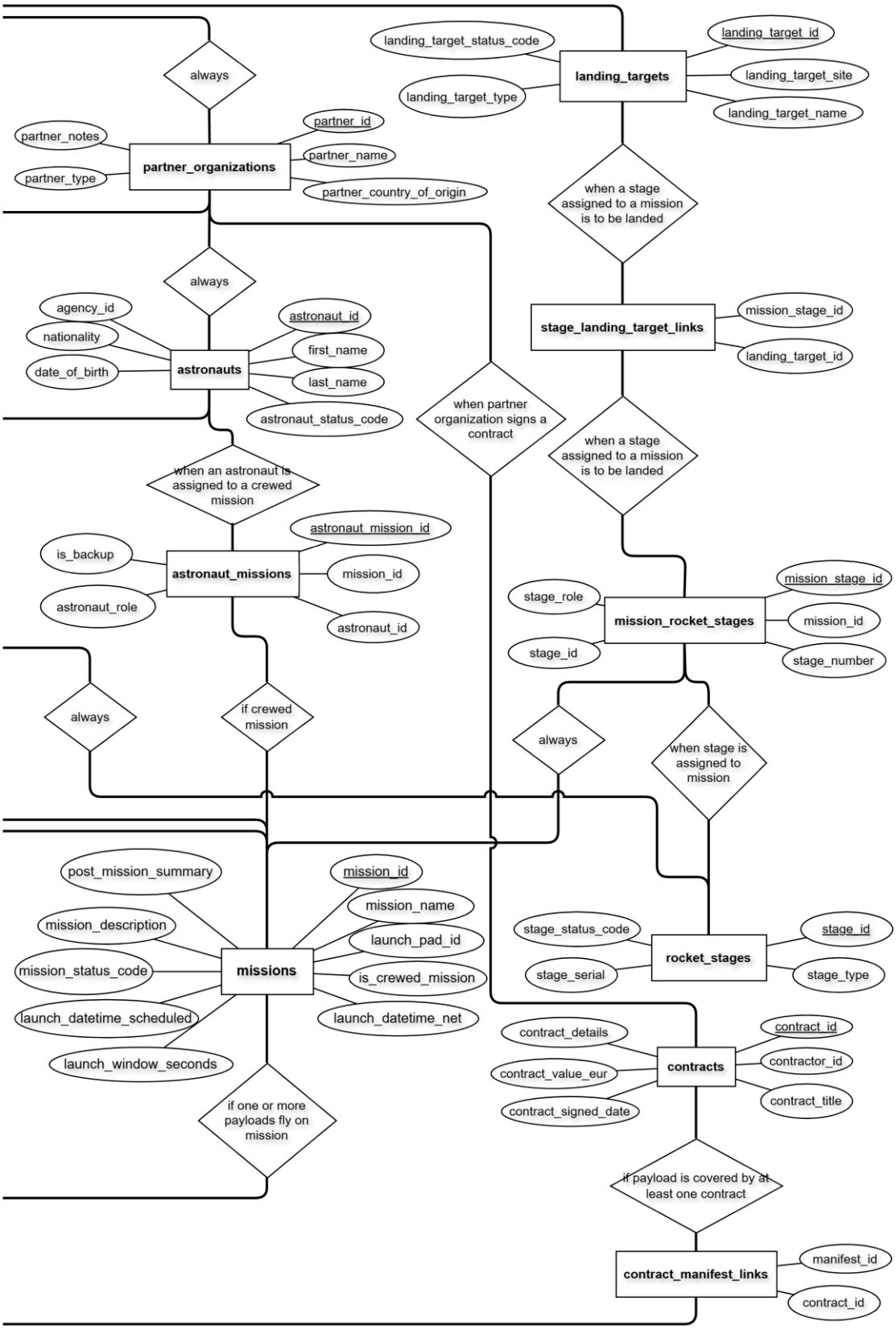
Küldetés azonosítója, küldetés neve, indítóállás azonosítója, emberes küldetés-e?, indítás legkorábbi időpontja, indítási ablak hossza másodpercben, indítás tervezett időpontja, státuszkód a küldetésre vonatkozóan, küldetés leírása, utólagos küldetés összefoglaló, hasznos teher repülési jegyzékének azonosítója, szerződés azonosítója, hasznos teher repülési jegyzékének azonosítója, hasznos teher azonosítója, célpálya azonosítója, hasznos teher leválasztásának (vagyis "üzembe helyezésének") státuszkódja, célpálya azonosítója, célpálya megnevezése, pálya apoapszisa (távolpontja) centiméter pontossággal, pálya periapszisa (közelpontja) centiméter pontossággal, pálya inklinációja (hajlásszöge) fokban, bolygóközi pálya-e?, cél égitest, megjegyzések a pályához, rakétafokozat küldetéshez rendelésének azonosítója, küldetés azonosítója, rakétafokozat azonosítója, rakétafokozat száma, rakétafokozat szerepe a küldetéshez használt rakéta konfigurációjában, rakétafokozat azonosítója, rakétafokozat típusa, rakétafokozat sorozatszáma, rakétafokozat státuszkódja, rakétafokozat küldetéshez rendelésének azonosítója, rakétafokozat visszatérési célpontjának azonosítója, rakétafokozat visszatérési célpontjának azonosítója, visszatérési célpont helyszíne, visszatérési célpont neve, visszatérési célpont típusa, visszatérési célpont státuszkódja, rakétaindítás helyszínének azonosítója, rakétaindítás helyszínének neve, rakátaindítás helyszínének országa, indítóállás azonosítója, rakétaindítás helyszínének

azonosítója, indítóállás neve, indítóállás pozíciója (földrajzi hosszúsága), indítóállás pozíciója (földrajzi szélessége), indítóállás státuszkódja, hasznos teher (rakomány) azonosítója, hasznos teher sorozatszáma, hasznos teher neve, hasznos teher típusa, hasznos teher tömege grammban, hasznos teher gyártójának azonosítója, hasznos teher megrendelőjének azonosítója, hasznos teher státuszkódja, űrhajós küldetéshez rendelésének azonosítója, küldetés azonosítója, űrhajós azonosítója, űrhajós szerepe a küldetésen, tartalék űrhajós-e?, űrhajós (általánosítva) azonosítója, űrhajós keresztneve, űrhajós családneve, űrhajós születésének napja, űrhajós nemzetisége, űrhajóst alkalmazó szervezet azonosítója, űrhajós státuszkódja, szerződés azonosítója, szerződéskötő szervezet azonosítója, szerződés címe, szerződéskötés dátuma, szerződés értéke euróban, szerződés részletei, partner szervezet azonosítója, partner szervezet neve, partner szervezet típusa, partner szervezet származási országa, megjegyzések a partner szervezethez, státuszkód, státusz neve ui-ban való megjelenítéshez, státuszkód, státusz kategória kódja, státusz kategória kódja, státusz kategória neve ui-ban való megjelenítéshez és szűréshez.

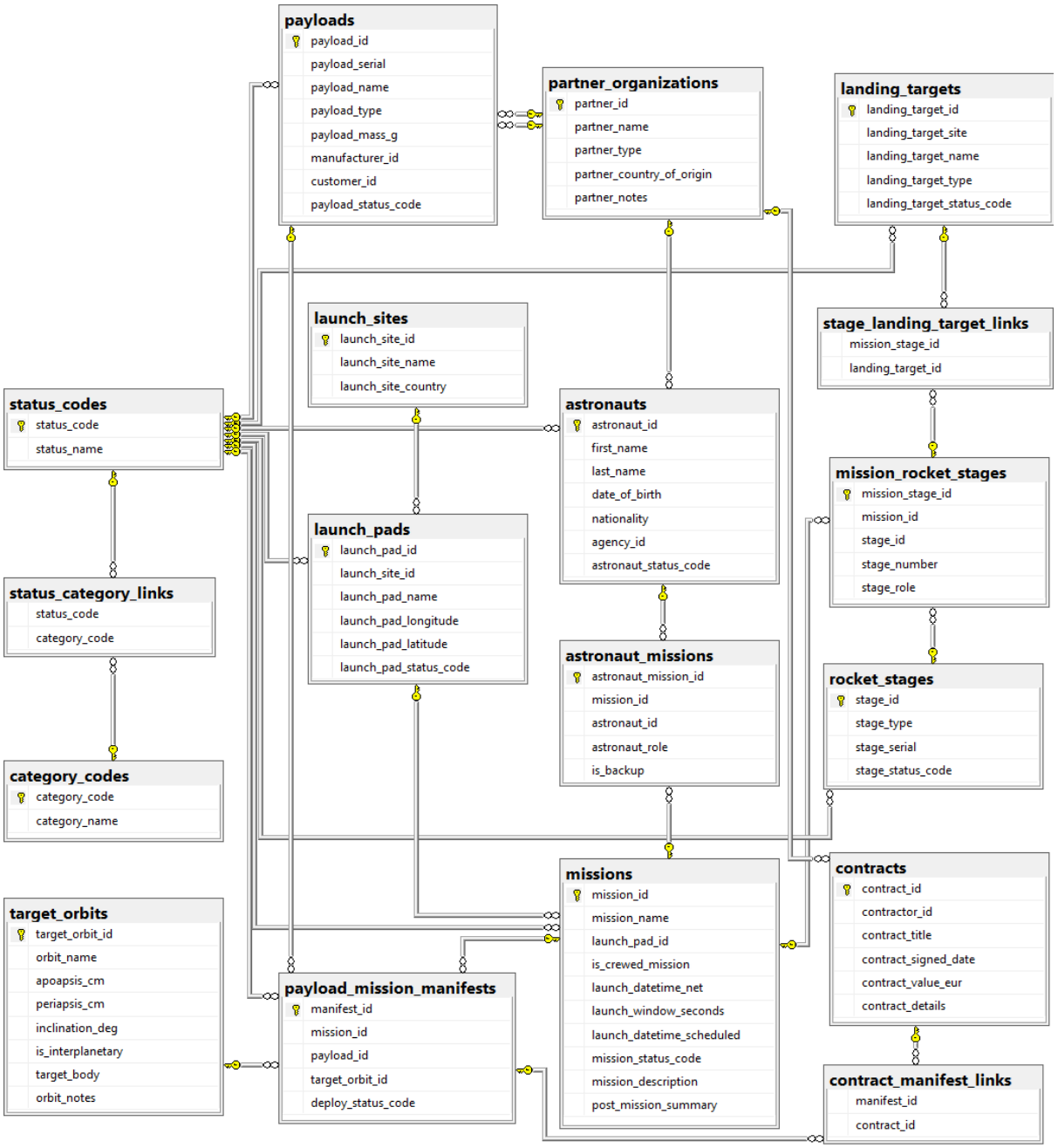
2. Egyed-kapcsolat diagram

Az olvashatóság megőrzése érdekében a diagram két részben kerül beillesztésre.




landing_target_status_code
landing_target_id
landing_target_site
landing_target_name
landing_target_type
landing_targets
always
partner_notes
partner_id
partner_name
partner_type
partner_organizations
partner_country_of_origin
when a stage assigned to a mission is to be landed
always
agency_id
nationality
date_of_birth
astronauts
astronaut_id
first_name
last_name
astronaut_status_code
stage_landing_target_links
mission_stage_id
landing_target_id
when partner organization signs a contract
when a stage assigned to a mission is to be landed
when an astronaut is assigned to a crewed mission
astronaut_mission_id
is_backup
astronaut_missions
mission_id
astronaut_role
astronaut_id
stage_role
mission_stage_id
mission_rocket_stages
mission_id
stage_id
stage_number
always
if crewed mission
always
when stage is assigned to mission
post_mission_summary
mission_id
mission_name
mission_description
launch_pad_id
stage_status_code
stage_id
mission_status_code
missions
is_crewed_mission
rocket_stages
stage_serial
stage_type
launch_datetime_scheduled
launch_datetime_net
contract_details
contract_id
launch_window_seconds
contracts
contract_value_eur
contractor_id
contract_signed_date
contract_title
if one or more payloads fly on mission
if payload is covered by at least one contract
contract_manifest_links
manifest_id
contract_id

3. Adatmodell bemutatása



4. Relációs sémák

- missions (mission_id, mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_window_seconds, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)
- contract_manifest_links (manifest_id, contract_id)
- payload_mission_manifests (manifest_id, payload_id, target_orbit_id, deploy_status_code)
- target_orbits (target_orbit_id, orbit_name, apoapsis_cm, periapsis_cm, inclination_deg, is_interplanetary, target_body)
- orbit_notes (mission_rocket_stages, mission_stage_id, mission_id, stage_id, stage_number, stage_role)
- rocket_stages (stage_id, stage_type, stage_serial, stage_status_code, stage_landing_target_links, mission_stage_id, landing_target_id)
- landing_targets (landing_target_id, landing_target_site, landing_target_name, landing_target_type, landing_target_status_code)
- launch_sites (launch_site_id, launch_site_name, launch_site_country)
- launch_pads (launch_pad_id, launch_site_id, launch_pad_name, launch_pad_longitude, launch_pad_latitude, launch_pad_status_code)
- payloads (payload_id, payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)
- astronaut_missions (astronaut_mission_id, mission_id, astronaut_id, astronaut_role, is_backup)
- astronauts (astronaut_id, first_name, last_name, date_of_birth, nationality, agency_id, astronaut_status_code)
- contracts (contract_id, contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details)
- partner_organizations (partner_id, partner_name, partner_type, partner_country_of_origin, partner_notes)
- status_codes (status_code, status_name)
- status_category_links (status_code, category_code)
- status_categories (category_code, category_name)

5. Adatmodell egyedeinek táblázatos bemutatása

5.1. missions

A küldetések adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni, vagy sem	Egy mondatos leírás.
mission_id	smallint	NOT NULL	Küldetés azonosítója
mission_name	nvarchar(50)	NOT NULL	Küldetés neve
launch_pad_id	smallint	NOT NULL	Indítóállás azonosítója
is_crewed_mission	bit	NOT NULL	Emberes küldetés-e?
launch_datetime_net	datetime	NOT NULL	Indítás legkorábbi időpontja ("no earlier than")
launch_window_seconds	int	NULL	Indítási ablak hossza másodpercben
launch_datetime_scheduled	datetime	NULL	Indítás tervezett időpontja
mission_status_code	tinyint	NOT NULL	Státuszkód a küldetésre vonatkozóan
mission_description	nvarchar(max)	NOT NULL	Küldetés leírása
post_mission_summary	nvarchar(max)	NULL	Utólagos küldetés összefoglaló
gyarto_id	int	NOT NULL	Gyártó azonosítója
gyarto_nev	nvarchar(50)	NOt NULL	Gyártó neve

5.2. contract_manifest_links

Egy vagy több hasznos terhet egy vagy több szerződéshez rendelő összekötő tábla, szükségessége a többször repülő, és/vagy több szerződés által tartalmazott rakományok miatt indokolt.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
manifest_id	int	NOT NULL	Hasznos teher repülési jegyzékének azonosítója
contract_id	smallint	NOT NULL	Szerződés azonosítója

5.3. payload_mission_manifests

Egy hasznos teher repülési jegyzékét tartalmazó tábla, szükségessége a több műholdat szállító („rideshare”) repülések miatt indokolt.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
manifest_id	int	NOT NULL	Hasznos teher repülési jegyzékének azonosítója
payload_id	int	NOT NULL	Hasznos teher azonosítója
target_orbit_id	int	NOT NULL	Célpálya azonosítója
deploy_status_code	tiny	NOT NULL	Hasznos teher leválasztásának (vagyis "üzembe helyezésének") státuszkódja

5.4. target_orbits

Hasznos teher célpályájának adatait tartalmazó tábla, a legfelső fokozat erre a pályára szállítja az adott rakományt.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
target_orbit_id	int	NOT NULL	Célpálya azonosítója
orbit_name	varchar(25)	NOT NULL	Célpálya megnevezése
apoapsis_cm	bigint	NOT NULL	Pálya apoapszisa (távolpontja) centiméter pontossággal
periapsis_cm	bigint	NOT NULL	Pálya periapszisa (közelpontja) centiméter pontossággal
inclination_deg	decimal(6,3)	NOT NULL	Pálya inklinációja (hajlásszöge) fokban
is_interplanetary	bit	NOT NULL	Bolygóközi pálya? - IGAZ esetben apoapsis_cm, periapsis_cm, inclination_deg mezők a Naphoz relatívan értendők
target_body	varchar(25)	NULL	Cél égitest - is_interplanetary = IGAZ esetén megadandó
orbit_notes	varchar(4000)	NOT NULL	Megjegyzések a pályához

5.5. mission_rocket_stages

Küldetéshez hozzárendelt rakétafokozat adatait tartalmazó tábla. Szükségességét a különböző típusú és szerepű fokozatok egységes kezelése indokolja.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
mission_stage_id	smallint	NOT NULL	Rakétafokozat küldetéshez rendelésének azonosítója
mission_id	smallint	NOT NULL	Küldetés azonosítója
stage_id	smallint	NOT NULL	Rakétafokozat azonosítója
stage_number	tinyint	NOT NULL	Rakétafokozat száma
stage_role	varchar(25)	NOT NULL	Rakétafokozat szerepe a küldetéshez használt rakéta konfigurációjában

5.6. rocket_stages

Rakétafokozatok adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
stage_id	smallint	NOT NULL	Rakétafokozat azonosítója
stage_type	varchar(25)	NOT NULL	Rakétafokozat típusa (pl: "Excelsior 1 Stage 1")
stage_serial	varchar(50)	NOT NULL	Rakétafokozat sorozatszáma [1]
stage_status_code	tinyint	NOT NULL	Rakétafokozat státuszkódja

[1] A sorozatszám egyedi, de nem ez az elsődleges kulcs, mivel a való életben a küldetések között a megszokott módtól eltérően változhat. Erre jó példa a Falcon 9 első fokozatának sorozatszáma (v1.1-től): B1*SN-MN* ahol *SN* a (három jegyű) sorszám és *MN* az (első fokozat szempontjából) teljesített küldetések száma. Például: B1067-31; ennek jelentése B = „booster”, 1 = első fokozat, 067 = sorszám és 31 = repülések száma (ezen sorok írásakor).

5.7. stage_landing_target_links

Rakétafokozatok landolási helyét és küldetéshez rendelt rakétafokozatokat összekötő tábla. Szükségességét a (célpálya energiaigénye miatt) nem visszatérő fokozatok esetében, az visszatérési hely idegen kulcsának nullozhatóságának, vagy „null értékű” visszatérési hely tábla csatolásának kiváltása indokolja.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
mission_stage_id	smallint	NOT NULL	Rakétafokozat küldetéshez rendelésének azonosítója
landing_target_id	smallint	NOT NULL	Rakétafokozat visszatérési célpontjának azonosítója

5.8. landing_targets

Rakétafokozatok visszatérési helyének adatait tartalmazó tábla. Az indítóállástól vett távolság és irány jelen esetben nem tekinthető küldetésirányítási problémának.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
landing_target_id	smallint	NOT NULL	Rakétafokozat visszatérési célpontjának azonosítója
landing_target_site	nvarchar(50)	NULL	Visszatérési célpont helyszíne [1]
landing_target_name	varchar(25)	NOT NULL	Visszatérési célpont neve
landing_target_type	varchar(15)	NOT NULL	Visszatérési célpont típusa
landing_target_status_code	tinyint	NOT NULL	Visszatérési célpont státuszkódja

[1]: A vízre (drónhajóra) történő visszatérés esetében nem adható meg helyszín név, csak a drónhajó neve.

5.9. launch_sites

Az indítási helyszínek adatait tartalmazó tábla. Egy indítási helyszínhez tartozhat több indítóállás (például Kennedy Space Center, Florida, USA; itt több vállalat több indítóállást is használ)

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
launch_site_id	smallint	NOT NULL	Rakétaindítás helyszínének azonosítója
launch_site_name	nvarchar(25)	NOT NULL	Rakétaindítás helyszínének neve
launch_site_countr y	varchar(25)	NOT NULL	Rakátaindítás helyszínének országa

5.10. launch_pads

Az indítóállás adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
launch_pad_id	smallint	NOT NULL	Indítóállás azonosítója
launch_site_id	smallint	NOT NULL	Rakétaindítás helyszínének azonosítója
launch_pad_name	varchar(25)	NOT NULL	Indítóállás neve
launch_pad_longitude	decimal(9,6)	NOT NULL	Indítóállás pozíciója (földrajzi hosszúsága)
launch_pad_latitude	decimal(9,6)	NOT NULL	Indítóállás pozíciója (földrajzi szélessége)
launch_pad_status_co de	tinyint	NOT NULL	Indítóállás státuszkódja

5.11. payloads

Egy hasznos teher adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
payload_id	int	NOT NULL	Hasznos teher (rakomány) azonosítója
payload_serial	varchar(50)	NOT NULL	Hasznos teher sorozatszáma
payload_name	varchar(50)	NOT NULL	Hasznos teher neve
payload_type	varchar(25)	NOT NULL	Hasznos teher típusa
payload_mass_g	int	NOT NULL	Hasznos teher tömege grammban
manufacturer_id	smallint	NOT NULL	Hasznos teher gyártójának azonosítója
customer_id	smallint	NOT NULL	Hasznos teher megrendelőjének azonosítója
payload_status_code	tinyint	NOT NULL	Hasznos teher státuszkódja

5.12. astronaut_missions

Egy űrhajós egy küldetésének adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
astronaut_mission_id	smallint	NOT NULL	Űrhajós küldetéshez rendelésének azonosítója
mission_id	smallint	NOT NULL	Küldetés azonosítója
astronaut_id	smallint	NOT NULL	Űrhajós azonosítója
astronaut_role	varchar(50)	NOT NULL	Űrhajós szerepe a küldetésen
is_backup	bit	NOT NULL	Tartalék űrhajós-e?

5.13. astronauts

Űrhajósok adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
astronaut_id	smallint	NOT NULL	Űrhajós (általánosítva) azonosítója
first_name	nvarchar(100)	NOT NULL	Űrhajós keresztneve
last_name	nvarchar(100)	NOT NULL	Űrhajós családneve
date_of_birth	date	NOT NULL	Űrhajós születésének napja
nationality	varchar(50)	NOT NULL	Űrhajós nemzetisége
agency_id	smallint	NOT NULL	Űrhajóst alkalmazó szervezet azonosítója
astronaut_status_code	tinyint	NOT NULL	Űrhajós státuszkódja

5.14. contracts

Szerződések adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
contract_id	smallint	NOT NULL	Szerződés azonosítója
contractor_id	smallint	NOT NULL	Szerződéskötő szervezet azonosítója
contract_title	varchar(50)	NOT NULL	Szerződés címe
contract_signed_date	date	NOT NULL	Szerződéskötés dátuma
contract_value_eur	decimal(18,2)	NOT NULL	Szerződés értéke Euróban
contract_details	nvarchar(max)	NOT NULL	Szerződés részletei

5.15. partner_organizations

Partner szervezetek adatait tartalmazó tábla.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
partner_id	smallint	NOT NULL	Partner szervezet azonosítója
partner_name	nvarchar(50)	NOT NULL	Partner szervezet neve
partner_type	varchar(25)	NOT NULL	Partner szervezet típusa
partner_country_of_origin	varchar(25)	NOT NULL	Partner szervezet származási országa
partner_notes	nvarchar(300)	NOT NULL	Megjegyzések a partner szervezethez

5.16. status_codes

Státusz kódokat és a hozzájuk tartozó státusz neveket tartalmazó tábla. Szükségességét a státuszok egységessége és UI-ban egyszerűbb kezelhetősége indokolja.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
status_code	tinyint	NOT NULL	Státuszkód
status_name	varchar(25)	NOT NULL	Státusz neve UI-ban való megjelenítéshez

5.17. status_category_links

Státuszokat és státusz kategóriákat összekötő tábla. Szükségességét a különböző, nem mindenhez hozzárendelhető státuszok szűrése indokolja.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
status_code	tinyint	NOT NULL	Státuszkód
category_code	tinyint	NOT NULL	Státusz kategória kódja

5.18. status_categories

Státusz kategóriákat tartalmazó tábla. Szükségességét a különböző, nem mindenhez hozzárendelhető státuszok szűrése és UI-ban egyszerűbb kezelhetősége indokolja.

Tulajdonság neve	Adattípusa (hossza)	Kötelező-e megadni,	Tulajdonság neve
category_code	tinyint	NOT NULL	Státusz kategória kódja
category_name	varchar(25)	NOT NULL	Státusz kategória neve UI-ban való megjelenítéshez és szűréshez

6. Megszorítási szabályok

6.1. astronauts

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
date_of_birth	date	DATEDIFF(YEAR, date_of_birth, GETDATE()) >= 21	Az űrhajósnak legalább 21 évesnek kell lennie.

6.2. contracts

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
contract_signed_date	date	contract_signed_date > DATEADD(MONTH, -1, GETDATE())	A szerződés aláírásának dátuma nem lehet egy hónapnál régebbi.
contract_value_eur	decimal(18,2)	contract_value_eur >= 0	A szerződés értéke nem lehet negatív.

6.3. target_orbits

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
(is_interplanetary, target_body)	bit, varchar(25)	(is_interplanetary = 1 AND target_body IS NOT NULL) OR (is_interplanetary = 0 AND target_body IS NULL) OR (is_interplanetary = 0 AND target_body = 'Earth')	A cél-égitest (target_body) csak akkor adható meg, ha a pálya bolygóközi (is_interplanetary = IGAZ).
(apoapsis_cm, periapsis_cm)	bigint, bigint	apoapsis_cm >= periapsis_cm	A pályamagasság távolpontja (apoapsis_cm)

			nem lehet kisebb, mint a közelpontja (periapsis_cm).
inclination_deg	decimal(6,3)	inclination_deg BETWEEN 0 AND 180	A pályahajlás (inclination_deg) 0 és 180 fok között kell, hogy legyen.

6.4. launch_pads

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
launch_pad_latitude	decimal(9,6)	launch_pad_latitude BETWEEN -90 AND 90	A földrajzi szélesség (launch_pad_latitude) -90 és 90 fok között kell, hogy legyen.
launch_pad_longitude	decimal(9,6)	launch_pad_longitude BETWEEN -180 AND 180	A földrajzi hosszúság (launch_pad_longitude) -180 és 180 fok között kell, hogy legyen.

6.5. missions

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
launch_window_seconds	int	(launch_window_seconds IS NULL) OR (launch_window_seconds >= 0)	Az indítási ablak (launch_window_seconds) nem lehet negatív, de lehet NULL (nem meghatározott).
launch_datetime_net	datetime	launch_datetime_net > GETDATE()	A legkorábbi indítási időpont (launch_datetime_net) csak jövőbeli dátum lehet.
launch_datetime_scheduled	datetime	(launch_datetime_scheduled IS NULL) OR (launch_datetime_scheduled > GETDATE())	A tervezett indítási időpont (launch_datetime_scheduled) jövőbeli dátum kell, hogy legyen, vagy

			lehet NULL.

6.6. mission_rocket_stages

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
stage_number	tinyint	stage_number BETWEEN 1 AND 9	A rakétafokozat sorszáma (stage_number) 1 és 9 közötti érték kell, hogy legyen.

6.7. payloads

Tulajdonság neve	Adattípusa (hossza)	Megszorítási szabály	Validálás szöveges értelmezése.
payload_mass_g	int	payload_mass_g > 0	A hasznos teher tömege (payload_mass_g) csak pozitív lehet.

7. Elsődleges kulcsok

- missions - mission_id
- target_orbits - target_orbit_id
- payload_mission_manifests - manifest_id
- mission_rocket_stages - mission_stage_id
- rocket_stages - stage_id
- landing_targets - landing_target_id
- launch_sites - launch_site_id
- launch_pads - launch_pad_id
- payloads - payload_id
- astronaut_missions - astronaut_mission_id
- astronauts - astronaut_id
- contracts - contract_id
- partner_organizations - partner_id
- status_codes - status_code
- status_categories - category_code

8. Indexek

- missions - mission_id
- missions - mission_name
- missions - launch_pad_id
- missions - is_crewed_mission
- missions - launch_datetime_net
- missions - mission_status_code
- contract_manifest_links - manifest_id
- contract_manifest_links - contract_id
- payload_mission_manifests - manifest_id
- payload_mission_manifests - payload_id
- payload_mission_manifests - target_orbit_id
- payload_mission_manifests - deploy_status_code
- target_orbits - target_orbit_id
- target_orbits - orbit_name
- mission_rocket_stages - mission_stage_id
- mission_rocket_stages - mission_id
- mission_rocket_stages - stage_id
- rocket_stages - stage_id
- rocket_stages - stage_type
- rocket_stages - stage_serial
- rocket_stages - stage_status_code
- stage_landing_target_links - mission_stage_id
- stage_landing_target_links - landing_target_id
- landing_targets - landing_target_id
- landing_targets - landing_target_name
- landing_targets - landing_target_status_code
- launch_sites - launch_site_id

- launch_sites - launch_site_name
- launch_sites - launch_site_country
- launch_pads - launch_pad_id
- launch_pads - launch_site_id
- launch_pads - launch_pad_name
- launch_pads - launch_pad_status_code
- payloads - payload_id
- payloads - payload_name
- payloads - manufacturer_id
- payloads - customer_id
- payloads - payload_status_code
- astronaut_missions - astronaut_mission_id
- astronaut_missions - mission_id
- astronaut_missions - astronaut_id
- astronauts - astronaut_id
- astronauts - first_name
- astronauts - last_name
- astronauts - agency_id
- astronauts - astronaut_status_code
- contracts - contract_id
- contracts - contractor_id
- contracts - contract_title
- partner_organizations - partner_id
- partner_organizations - partner_name
- status_codes - status_code
- status_codes - status_name
- status_category_links - status_code
- status_category_links - category_code
- status_categories - category_code
- status_categories - category_name

9. Alapértelmezett értékek

- missions - is_crewed_mission - FALSE
- missions - launch_window_seconds - 0
- target_orbits - is_interplanetary - FALSE

10. Automatikus növekmények

- missions - mission_id - IDENTITY(1,1)
- payload_mission_manifests - manifest_id - IDENTITY(1,1)
- target_orbits - target_orbit_id - IDENTITY(1,1)
- mission_rocket_stages - mission_stage_id - IDENTITY(1,1)
- rocket_stages - stage_id - IDENTITY(1,1)
- landing_targets - landing_target_id - IDENTITY(1,1)
- launch_sites - launch_site_id - IDENTITY(1,1)
- launch_pads - launch_pad_id - IDENTITY(1,1)
- payloads - payload_id - IDENTITY(1,1)

- astronaut_missions - astronaut_mission_id - IDENTITY(1,1)
- astronauts - astronaut_id - IDENTITY(1,1)
- contracts - contract_id - IDENTITY(1,1)
- partner_organizations - partner_id - IDENTITY(1,1)

11. Lekérdezések

11.1. Egytáblás lekérdezések

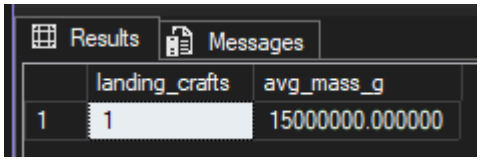
11.1.1. Adja meg a valaha szerződött leszállóegységek darabszámát és tömegük átlagát!

Megjegyzés: csak azok a hasznos terhek kerülnek az adatbázisba, amire a következő feltételek közül legalább az egyik igaz:

- van hozzárendelt szerződés vagy küldetés
- cégen belüli hasznos teher (pl tömegszimulátor, új űrkapszula)

```
use [Avantvia Mission Control Database]


select count(*) as landing_crafts, avg(cast(payload_mass_g as decimal(10,2)))
as avg_mass_g
from payloads
where payload_type = 'Lander'
```



11.1.2. Adja meg az aláírt szerződések darabszámát és összértékét pénzügyi évek szerint csoportosítva!

```
use [Avantvia Mission Control Database]


declare @fy_start_month tinyint = 4;


select
    -- if the month is less than the start month, it belongs to the fiscal year
started in the previous calendar year
    case
        when month(contract_signed_date) >= @fy_start_month
        then year(contract_signed_date)
        else year(contract_signed_date) - 1
    end as fiscal_year,
    count(*) as contract_count,
    sum(contract_value_eur) as total_value_eur
from contracts
group by
    case
```

```
            when month(contract_signed_date) >= @fy_start_month
            then year(contract_signed_date)
            else year(contract_signed_date) - 1
        end
    order by fiscal_year desc;
```

Results | Messages

	fiscal_year	contract_count	total_value_eur
1	2025	1	4000000000.00
2	2024	17	9100000.00
3	2023	23	3550000000.00
4	2022	8	750000000.00
5	2019	25	1250000000.00
6	2018	1	2600000000.00
7	2014	23	115000000.00

11.2. Kéttáblás lekérdezések

11.2.1. Listázza ki az összes indítóállást helyszín szerint csoportosítva!

```
use [Avantvia Mission Control Database]


select launch_pads.launch_pad_name, launch_sites.launch_site_name
from launch_pads
   inner join launch_sites on
launch_pads.launch_site_id=launch_sites.launch_site_id
order by launch_sites.launch_site_id
```

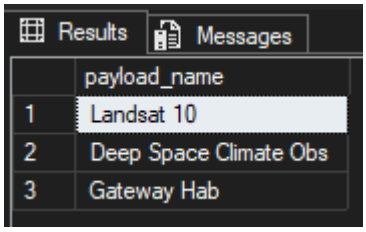
Results | Messages

	launch_pad_name	launch_site_name
1	Pad A	Cape Orizon
2	Pad B	Cape Orizon
3	Pad C	Cape Orizon
4	Pad D	Vandenberg SFB
5	Pad E	Guiana Space Centre

11.2.2. Listázza ki az összes NASA által megrendelt rakományt!

```
use [Avantvia Mission Control Database]


select payload_name
from payloads
   inner join partner_organizations on
payloads.customer_id=partner_organizations.partner_id
            and partner_organizations.partner_name = 'NASA'
```



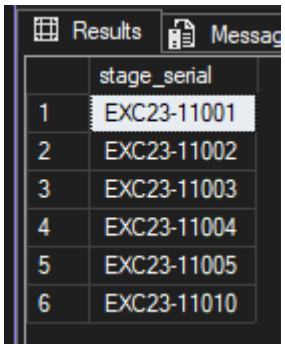
11.2.3. Melyik Excelsior 2 első fokozatok állnak rendelkezésre?

Megjegyzések:

- Az Excelsior rakétacsalád szériaszáma a következő mintát követi: `EXC[Variáns][Generáció]-[Fokozatszám][Sorszám]`, ahol:
  - Variáns: 1, 2, N (Nova)
  - Generáció: 1 jegyű szám
  - Fokozatszám: 1, 2, K (kickstage)
  - Sorszám: 4 jegyű gyártási sorszám
- Tisztában vagyok vele, hogy ez technikailag megsérti a normálformát, de a sorozatszámok változásának sokkal kisebb az esélye, mint annak, hogy az *emberek* hogyan nevezik el a fokozatot a `rocket_stages.stage_type` oszlopban

```
use [Avantvia Mission Control Database]


select stage_serial
from rocket_stages
   inner join status_codes on
rocket_stages.stage_status_code=status_codes.status_code and
status_codes.status_name = 'Ready' and stage_serial like 'EXC2_-1%'
```



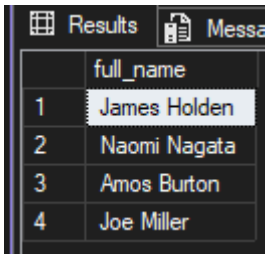
11.2.4. Listázza a jelenleg aktív és szabad űrhajósokat!

Csak olyan űrhajós kerül az adatbázisba, akihez rendelhető szerződés vagy küldetés.

```
use [Avantvia Mission Control Database]


select first_name + ' ' + last_name as full_name
from astronauts
```

```
    inner join status_codes on
status_codes.status_code=astronauts.astronaut_status_code and
status_codes.status_name = 'Ready'
```



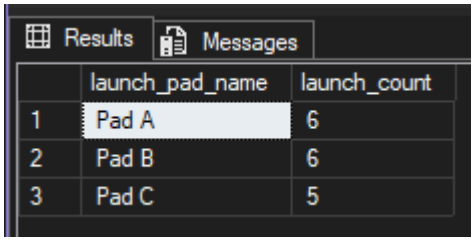
Results | Messa

	full_name
1	James Holden
2	Naomi Nagata
3	Amos Burton
4	Joe Miller

11.2.5. Listázza a 3 legtöbbet használt indítóállást, és az onnan végrehajtott indítások számát az utóbbi évben!

```
use [Avantvia Mission Control Database]


select top 3 with ties
    launch_pads.launch_pad_name, count(missions.mission_id) as launch_count
from launch_pads
    inner join missions on missions.launch_pad_id=launch_pads.launch_pad_id
where datediff(day, missions.launch_datetime_scheduled, getdate()) between 0
and 365
group by launch_pads.launch_pad_name
order by launch_count desc
```



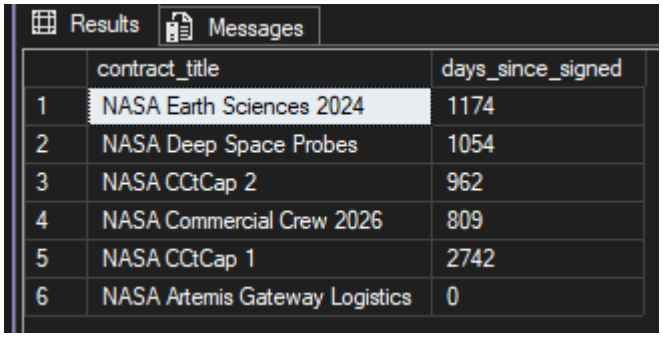
Results | Messages

	launch_pad_name	launch_count
1	Pad A	6
2	Pad B	6
3	Pad C	5

11.2.6. Listázza az összes NASA szerződést, és hogy hány napja írták ezeket alá!

```
use [Avantvia Mission Control Database]


select contract_title, datediff(day, contract_signed_date, getdate()) as
days_since_signed
from contracts
    inner join partner_organizations on
contracts.contractor_id=partner_organizations.partner_id and partner_name = 'NASA'
```



Results | Messages

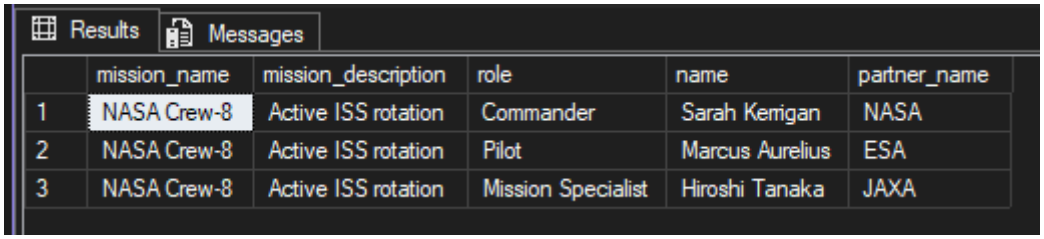
	contract_title	days_since_signed
1	NASA Earth Sciences 2024	1174
2	NASA Deep Space Probes	1054
3	NASA CCtCap 2	962
4	NASA Commercial Crew 2026	809
5	NASA CCtCap 1	2742
6	NASA Artemis Gateway Logistics	0

11.3. Többtáblás lekérdezések

11.3.1. Listázza a jelenleg folyamatban lévő emberes küldetés(ek) adatait: küldetés neve, leírása, űrhajósok teljes neve, szerepköreik, szervezetek

```
use [Avantvia Mission Control Database]


select
    missions.mission_name, missions.mission_description,
    case when astronaut_missions.is_backup = 0 then
astronaut_missions.astronaut_role
when astronaut_missions.is_backup = 1 then 'Backup ' +
astronaut_missions.astronaut_role
end as 'role',
    astronauts.first_name + ' ' + astronauts.last_name as 'name',
partner_organizations.partner_name
from missions
    inner join status_codes on
missions.mission_status_code=status_codes.status_code and
missions.is_crewed_mission=1
    and status_codes.status_name not in ('Under Planning', 'Under Analysis',
'Cancelled', 'Awaiting', 'Success', 'Failure', 'Partial Failure', 'Scrub')
    inner join astronaut_missions on
missions.mission_id=astronaut_missions.mission_id
    inner join astronauts on
astronaut_missions.astronaut_id=astronauts.astronaut_id
    inner join partner_organizations on
astronauts.agency_id=partner_organizations.partner_id
```



Results | Messages

	mission_name	mission_description	role	name	partner_name
1	NASA Crew-8	Active ISS rotation	Commander	Sarah Kerrigan	NASA
2	NASA Crew-8	Active ISS rotation	Pilot	Marcus Aurelius	ESA
3	NASA Crew-8	Active ISS rotation	Mission Specialist	Hiroshi Tanaka	JAXA

11.3.2. Listázza ki a következő rideshare küldetés adatait: küldetés neve, legkorábbi indítási dátuma, rakományok neve, gyártója, készültségi státusza, hozzárendelt pályák neve

A rideshare („küldetésmegosztó”) küldetések olyan küldetések, ahol több megrendelő rakománya repül egyszerre ugyan azon a rakétán.

```
use [Avantvia Mission Control Database]
go


with
    rideshare_missions
    as
    (
        select payload_mission_manifests.mission_id
        from payload_mission_manifests
            inner join payloads on payload_mission_manifests.payload_id = payloads.payload_id
        group by payload_mission_manifests.mission_id
        having count(distinct payloads.customer_id) > 1
    )
select distinct
    mission_name, launch_datetime_net, mission_status_name, payload_name, manufacturer_name, orbit_name, payload_status_name
from (select top 1 with ties
        missions.mission_name, missions.launch_datetime_net, mission_status.status_name as mission_status_name,
        payloads.payload_name, manufacturer.partner_name as manufacturer_name,
        target_orbits.orbit_name, payload_status.status_name as payload_status_name
    from missions
        inner join rideshare_missions on missions.mission_id = rideshare_missions.mission_id
        inner join status_codes as mission_status
        on missions.mission_status_code = mission_status.status_code and mission_status.status_name in ('Ready', 'Awaiting', 'To Be Determined', 'Under Planning')
        and (missions.launch_datetime_net >= getdate() or (mission_status.status_name = 'Scrub' and missions.launch_datetime_scheduled is null))
        inner join payload_mission_manifests
        on missions.mission_id = payload_mission_manifests.mission_id
        inner join payloads
        on payload_mission_manifests.payload_id = payloads.payload_id
```

```
            inner join status_codes as payload_status
            on payloads.payload_status_code = payload_status.status_code
            inner join partner_organizations as manufacturer
            on payloads.manufacturer_id = manufacturer.partner_id
            inner join partner_organizations as customer
            on payloads.customer_id = customer.partner_id
            inner join target_orbits
            on payload_mission_manifests.target_orbit_id =
target_orbits.target_orbit_id
      order by isnull(missions.launch_datetime_scheduled,
missions.launch_datetime_net) asc) as next_rideshare
   order by next_rideshare.mission_name
```

Results | Messages

	mission_name	launch_datetime_net	mission_status_name	payload_name	manufacturer_name	orbit_name	payload_status_name
1	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Academic Apex 9	MIT	SSO	Ready
2	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Academic Horizon 4	MIT	SSO	Ready
3	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Research Odin 2	MIT	SSO	Ready
4	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Research Polaris 10	MIT	SSO	Ready
5	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Research Vega 11	MIT	SSO	Ready
6	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Student Nyx 1	MIT	SSO	Ready
7	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Student Vanguard 8	MIT	SSO	Ready
8	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Student Zeus 5	MIT	SSO	Ready
9	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Tech Chronos 3	MIT	SSO	Ready
10	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Tech Horizon 7	MIT	SSO	Ready
11	Cosmic-Share 7	2026-11-15 14:00:00.000	Under Planning	Tech Sol 6	MIT	SSO	Ready

11.3.3. Listázza ki a következő rideshare küldetéshez tartozó szerződések adatait: küldetés neve, legkorábbi indítási dátuma, szerződések címe, értéke, partner neve

```
   use [Avantvia Mission Control Database]
   go


   with
      rideshare_missions
      as
      (
            select payload_mission_manifests.mission_id
            from payload_mission_manifests
                  inner join payloads on payload_mission_manifests.payload_id =
payloads.payload_id
            group by payload_mission_manifests.mission_id
            having count(distinct payloads.customer_id) > 1
      )
   select distinct mission_name, launch_datetime_net, contract_title,
contract_value_eur, partner_name
   from (select top 1 with ties
```

```
            missions.mission_name, missions.launch_datetime_net,
            contracts.contract_title, contracts.contract_value_eur,
            partner_organizations.partner_name
      from missions
            inner join rideshare_missions on missions.mission_id =
rideshare_missions.mission_id
            inner join status_codes as mission_status
            on missions.mission_status_code = mission_status.status_code and
mission_status.status_name in ('Ready', 'Awaiting', 'To Be Determined', 'Under
Planning')
            and (missions.launch_datetime_net >= getdate() or
(mission_status.status_name = 'Scrub' and missions.launch_datetime_scheduled is
null))
            inner join payload_mission_manifests
            on missions.mission_id = payload_mission_manifests.mission_id
            inner join contract_manifest_links on
payload_mission_manifests.manifest_id = contract_manifest_links.manifest_id
            inner join contracts on contract_manifest_links.contract_id =
contracts.contract_id
            inner join partner_organizations on contracts.contractor_id =
partner_organizations.partner_id
      order by isnull(missions.launch_datetime_scheduled,
missions.launch_datetime_net) asc) as next_rideshare
   order by next_rideshare.mission_name
```

Results | Messages

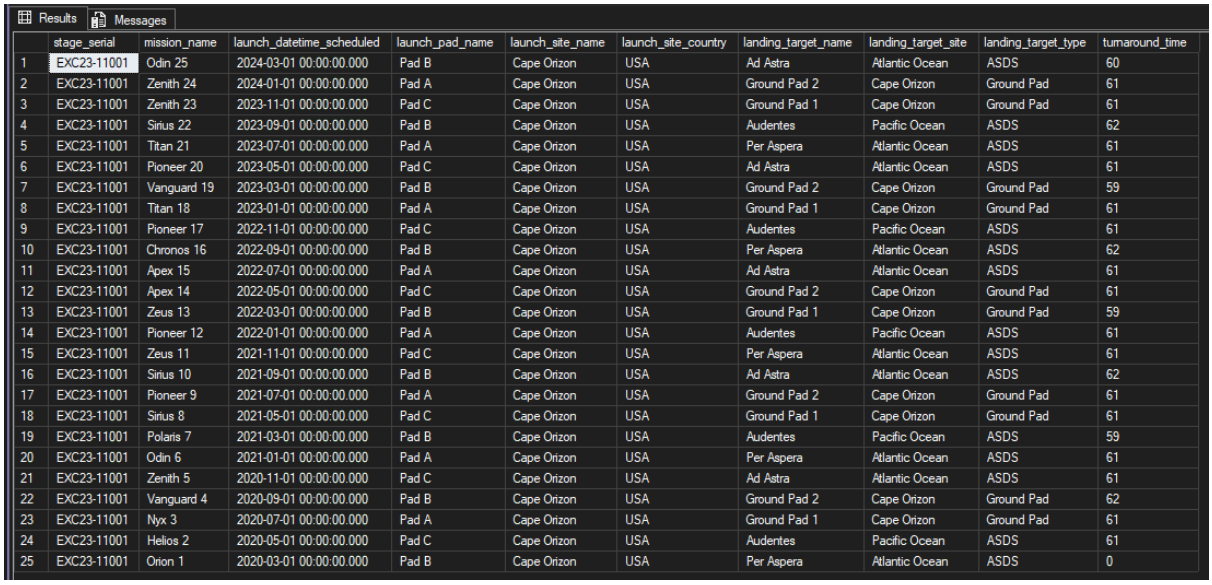
	mission_name	launch_datetime_net	contract_title	contract_value_eur	partner_name
1	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Apex 11	500000.00	Max Planck Institute
2	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Ares 5	500000.00	Max Planck Institute
3	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Helios 2	500000.00	Oxford
4	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Hermes 7	500000.00	Stanford
5	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Luna 9	500000.00	Caltech
6	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Nova 4	500000.00	University of Tokyo
7	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Nyx 1	500000.00	Stanford
8	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Polaris 3	500000.00	Caltech
9	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Titan 10	500000.00	University of Tokyo
10	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Titan 8	500000.00	Oxford
11	Cosmic-Share 7	2026-11-15 14:00:00.000	University Rideshare Zeus 6	500000.00	MIT

11.3.4. Listázza ki a legtöbbet repült Excelsior 2 első fokozat repülési történetét: küldetések nevei, indítási dátuma, helyszíne, visszatérés célpontja, és az előző küldetés óta eltelt napok számát.

```
use [Avantvia Mission Control Database]


select rocket_stages.stage_serial,
```

```
      missions.mission_name, missions.launch_datetime_scheduled,
      launch_pads.launch_pad_name, launch_sites.launch_site_name,
launch_sites.launch_site_country,
      landing_targets.landing_target_name, landing_targets.landing_target_site,
landing_targets.landing_target_type,
      isnull(datediff(day, lag(cast(missions.launch_datetime_scheduled as date))
over (partition by rocket_stages.stage_id order by
missions.launch_datetime_scheduled), missions.launch_datetime_scheduled), 0) as
[turnaround_time]
   from rocket_stages
      inner join
      (
   select top 1 with ties
            mission_rocket_stages.stage_id as stage_id
      from mission_rocket_stages
            inner join rocket_stages on rocket_stages.stage_id =
mission_rocket_stages.stage_id and rocket_stages.stage_serial like 'EXC2_-1%'
      group by mission_rocket_stages.stage_id
      order by count(mission_rocket_stages.mission_id) desc
   ) as most_exc2_flights on rocket_stages.stage_id = most_exc2_flights.stage_id
      inner join mission_rocket_stages on rocket_stages.stage_id =
mission_rocket_stages.stage_id
      inner join missions on mission_rocket_stages.mission_id =
missions.mission_id
      inner join launch_pads on missions.launch_pad_id =
launch_pads.launch_pad_id
      inner join launch_sites on launch_pads.launch_site_id =
launch_sites.launch_site_id
      left join stage_landing_target_links on
mission_rocket_stages.mission_stage_id =
stage_landing_target_links.mission_stage_id
      left join landing_targets on stage_landing_target_links.landing_target_id =
landing_targets.landing_target_id
   order by missions.launch_datetime_scheduled desc
```



Results | Messages

	stage_serial	mission_name	launch_datetime_scheduled	launch_pad_name	launch_site_name	launch_site_country	landing_target_name	landing_target_site	landing_target_type	turnaround_time
1	EXC23-11001	Odin 25	2024-03-01 00:00:00.000	Pad B	Cape Orizon	USA	Ad Astra	Atlantic Ocean	ASDS	60
2	EXC23-11001	Zenith 24	2024-01-01 00:00:00.000	Pad A	Cape Orizon	USA	Ground Pad 2	Cape Orizon	Ground Pad	61
3	EXC23-11001	Zenith 23	2023-11-01 00:00:00.000	Pad C	Cape Orizon	USA	Ground Pad 1	Cape Orizon	Ground Pad	61
4	EXC23-11001	Sirius 22	2023-09-01 00:00:00.000	Pad B	Cape Orizon	USA	Audentes	Pacific Ocean	ASDS	62
5	EXC23-11001	Titan 21	2023-07-01 00:00:00.000	Pad A	Cape Orizon	USA	Per Aspera	Atlantic Ocean	ASDS	61
6	EXC23-11001	Pioneer 20	2023-05-01 00:00:00.000	Pad C	Cape Orizon	USA	Ad Astra	Atlantic Ocean	ASDS	61
7	EXC23-11001	Vanguard 19	2023-03-01 00:00:00.000	Pad B	Cape Orizon	USA	Ground Pad 2	Cape Orizon	Ground Pad	59
8	EXC23-11001	Titan 18	2023-01-01 00:00:00.000	Pad A	Cape Orizon	USA	Ground Pad 1	Cape Orizon	Ground Pad	61
9	EXC23-11001	Pioneer 17	2022-11-01 00:00:00.000	Pad C	Cape Orizon	USA	Audentes	Pacific Ocean	ASDS	61
10	EXC23-11001	Chronos 16	2022-09-01 00:00:00.000	Pad B	Cape Orizon	USA	Per Aspera	Atlantic Ocean	ASDS	62
11	EXC23-11001	Apex 15	2022-07-01 00:00:00.000	Pad A	Cape Orizon	USA	Ad Astra	Atlantic Ocean	ASDS	61
12	EXC23-11001	Apex 14	2022-05-01 00:00:00.000	Pad C	Cape Orizon	USA	Ground Pad 2	Cape Orizon	Ground Pad	61
13	EXC23-11001	Zeus 13	2022-03-01 00:00:00.000	Pad B	Cape Orizon	USA	Ground Pad 1	Cape Orizon	Ground Pad	59
14	EXC23-11001	Pioneer 12	2022-01-01 00:00:00.000	Pad A	Cape Orizon	USA	Audentes	Pacific Ocean	ASDS	61
15	EXC23-11001	Zeus 11	2021-11-01 00:00:00.000	Pad C	Cape Orizon	USA	Per Aspera	Atlantic Ocean	ASDS	61
16	EXC23-11001	Sirius 10	2021-09-01 00:00:00.000	Pad B	Cape Orizon	USA	Ad Astra	Atlantic Ocean	ASDS	62
17	EXC23-11001	Pioneer 9	2021-07-01 00:00:00.000	Pad A	Cape Orizon	USA	Ground Pad 2	Cape Orizon	Ground Pad	61
18	EXC23-11001	Sirius 8	2021-05-01 00:00:00.000	Pad C	Cape Orizon	USA	Ground Pad 1	Cape Orizon	Ground Pad	61
19	EXC23-11001	Polaris 7	2021-03-01 00:00:00.000	Pad B	Cape Orizon	USA	Audentes	Pacific Ocean	ASDS	59
20	EXC23-11001	Odin 6	2021-01-01 00:00:00.000	Pad A	Cape Orizon	USA	Per Aspera	Atlantic Ocean	ASDS	61
21	EXC23-11001	Zenith 5	2020-11-01 00:00:00.000	Pad C	Cape Orizon	USA	Ad Astra	Atlantic Ocean	ASDS	61
22	EXC23-11001	Vanguard 4	2020-09-01 00:00:00.000	Pad B	Cape Orizon	USA	Ground Pad 2	Cape Orizon	Ground Pad	62
23	EXC23-11001	Nyx 3	2020-07-01 00:00:00.000	Pad A	Cape Orizon	USA	Ground Pad 1	Cape Orizon	Ground Pad	61
24	EXC23-11001	Helios 2	2020-05-01 00:00:00.000	Pad C	Cape Orizon	USA	Audentes	Pacific Ocean	ASDS	61
25	EXC23-11001	Orion 1	2020-03-01 00:00:00.000	Pad B	Cape Orizon	USA	Per Aspera	Atlantic Ocean	ASDS	0

11.3.5. Mennyi volt az előző pénzügyi évben a NASA küldetésekből származó összbevétel?

Pontosítás: a kérdés azokra a szerződésekre vonatkozik, amikhez tartozó indítások voltak az előző pénzügyi évben. Átfogalmazva, mennyi összbevételt befolyásoltak az előző pénzügyi félévben végrehajtott küldetések?

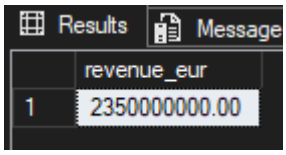
```
use [Avantvia Mission Control Database]


declare @fy_start_month tinyint
set @fy_start_month = 4


declare @current_fy_start date
set @current_fy_start = case
when month(getdate()) >= @fy_start_month then datefromparts(year(getdate()),
@fy_start_month, 1)
else datefromparts(year(getdate())-1, @fy_start_month, 1)
end


select sum(contract_value_eur) as revenue_eur
from contracts
where contracts.contract_id in (
select distinct contracts.contract_id
from contracts
    inner join partner_organizations on contracts.contractor_id =
partner_organizations.partner_id and partner_organizations.partner_name = 'NASA'
    inner join contract_manifest_links on contracts.contract_id =
contract_manifest_links.contract_id
    inner join payload_mission_manifests on contract_manifest_links.manifest_id
= payload_mission_manifests.manifest_id
```

```
      inner join missions on payload_mission_manifests.mission_id =
missions.mission_id
   where cast(isnull(missions.launch_datetime_scheduled, launch_datetime_net) as
date) between dateadd(year, -1, @current_fy_start) and dateadd(day, -1,
@current_fy_start) )
```

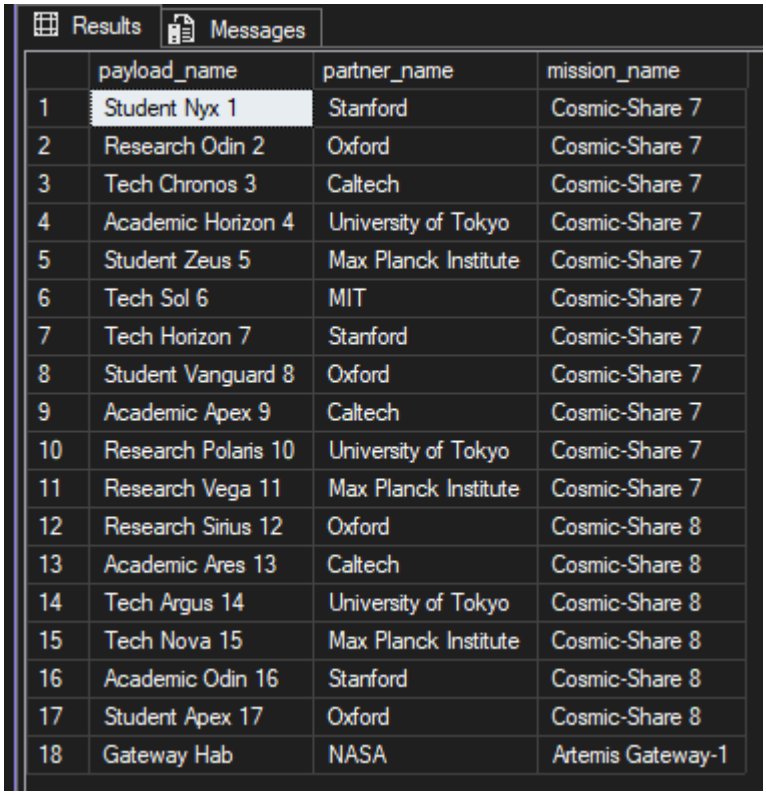


11.3.6. Listázza ki az összes „integrációra kész” rakomány adatait: rakomány neve, megrendelő neve, hozzárendelt küldetés neve.

A hasznos terhek esetében a ’Ready’ státusz azt jelenti, hogy készenáll a raktérbe szerelésre.

```
use [Avantvia Mission Control Database]


select payloads.payload_name,
   partner_organizations.partner_name,
   missions.mission_name
from payloads
   inner join status_codes on payloads.payload_status_code =
status_codes.status_code and status_codes.status_name = 'Ready'
   inner join partner_organizations on payloads.customer_id =
partner_organizations.partner_id
   inner join payload_mission_manifests on payloads.payload_id =
payload_mission_manifests.payload_id
   inner join missions on payload_mission_manifests.mission_id =
missions.mission_id
order by missions.launch_datetime_net asc
```



Results | Messages

	payload_name	partner_name	mission_name
1	Student Nyx 1	Stanford	Cosmic-Share 7
2	Research Odin 2	Oxford	Cosmic-Share 7
3	Tech Chronos 3	Caltech	Cosmic-Share 7
4	Academic Horizon 4	University of Tokyo	Cosmic-Share 7
5	Student Zeus 5	Max Planck Institute	Cosmic-Share 7
6	Tech Sol 6	MIT	Cosmic-Share 7
7	Tech Horizon 7	Stanford	Cosmic-Share 7
8	Student Vanguard 8	Oxford	Cosmic-Share 7
9	Academic Apex 9	Caltech	Cosmic-Share 7
10	Research Polaris 10	University of Tokyo	Cosmic-Share 7
11	Research Vega 11	Max Planck Institute	Cosmic-Share 7
12	Research Sirius 12	Oxford	Cosmic-Share 8
13	Academic Ares 13	Caltech	Cosmic-Share 8
14	Tech Argus 14	University of Tokyo	Cosmic-Share 8
15	Tech Nova 15	Max Planck Institute	Cosmic-Share 8
16	Academic Odin 16	Stanford	Cosmic-Share 8
17	Student Apex 17	Oxford	Cosmic-Share 8
18	Gateway Hab	NASA	Artemis Gateway-1

11.3.7. Adja meg az egyes visszatérési célpontok kihasználtságát az elmúlt évben.

Az építés alatt álló visszatérési célpontok nem kerülnek bele a listába, hiszen hozzájuk nem lehet visszatérést rendelni.

```
use [Avantvia Mission Control Database]


declare @current_date date
set @current_date = getdate()


select landing_targets.landing_target_name,
isnull(landing_targets.landing_target_site, '-') as landing_target_site,
landing_targets.landing_target_type,
count(stage_landing_target_links.mission_stage_id) as landings_assigned
from landing_targets
    left join stage_landing_target_links on landing_targets.landing_target_id =
stage_landing_target_links.landing_target_id
    left join mission_rocket_stages on
stage_landing_target_links.mission_stage_id =
mission_rocket_stages.mission_stage_id
    left join missions on mission_rocket_stages.mission_id =
missions.mission_id and cast(missions.launch_datetime_scheduled as date) between
dateadd(year, -1, @current_date) and @current_date
where landing_target_status_code not in (select status_code
from status_codes
where status_name = 'Under Construction')
```

```
   group by landing_targets.landing_target_name,
landing_targets.landing_target_site, landing_targets.landing_target_type

   order by landings_assigned desc
```

Results | Messages

	landing_target_name	landing_target_site	landing_target_type	landings_assigned
1	Ad Astra	Atlantic Ocean	ASDS	26
2	Ground Pad 1	Cape Orizon	Ground Pad	9
3	Ground Pad 2	Cape Orizon	Ground Pad	5
4	Audentes	Pacific Ocean	ASDS	5
5	Per Aspera	Atlantic Ocean	ASDS	5
6	Ground Pad 3	Vandenberg SFB	Ground Pad	0

11.3.8. Listázza ki a legértékesebb aktív szerződés adatait: szerződés címe, értéke, hozzárendelt küldetés neve, legkorábbi indítási dátuma, hozzárendelt hasznos terhek neve(i) és típusa(i)

```
   use [Avantvia Mission Control Database]


   select contracts.contract_title, contracts.contract_value_eur,
missions.mission_name, missions.launch_datetime_net, payloads.payload_name,
payloads.payload_type

   from contract_manifest_links

      inner join

      (select top 1 with ties

            contracts.contract_id

      from contracts

            inner join contract_manifest_links on contracts.contract_id =
contract_manifest_links.contract_id

            inner join payload_mission_manifests on
contract_manifest_links.manifest_id = payload_mission_manifests.manifest_id

            inner join missions on payload_mission_manifests.mission_id =
missions.mission_id and isnull(missions.launch_datetime_scheduled,
missions.launch_datetime_net) > getdate()

      group by contracts.contract_id, contracts.contract_value_eur

      order by contracts.contract_value_eur desc

   ) as most_valuable_contract on contract_manifest_links.contract_id =
most_valuable_contract.contract_id

      inner join contracts on most_valuable_contract.contract_id =
contracts.contract_id

      inner join payload_mission_manifests on contract_manifest_links.manifest_id
= payload_mission_manifests.manifest_id

      inner join payloads on payload_mission_manifests.payload_id =
payloads.payload_id

      inner join missions on payload_mission_manifests.mission_id =
missions.mission_id
```



Results | Messages

	contract_title	contract_value_eur	mission_name	launch_datetime_net	payload_name	payload_type
1	NASA Artemis Gateway Logistics	4000000000.00	Artemis Gateway-1	2027-03-20 02:35:42.183	Gateway Hab	Lander

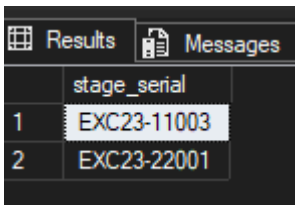
11.4. Alselectes lekérdezés

11.4.1. Listázza ki a legutóbbi küldetésen részt vett rakétafokozatok sorszámait.

Megjegyzés: az „elemzés alatt” státusz azt jelenti, hogy a küldetés befejeződött, de a mérnökök még az anomáliákkal és meghibásodásokkal kapcsolatos telemetriai adatok kiértékelését végzik.

```
use [Avantvia Mission Control Database]


select stage_serial
from rocket_stages
    inner join mission_rocket_stages on rocket_stages.stage_id =
mission_rocket_stages.stage_id
    inner join
    (select top 1 with ties
          mission_id, launch_datetime_net
    from missions
    where missions.mission_status_code in
    (select status_code
    from status_codes
    where status_name in ('Success', 'Failure', 'Partial Failure', 'Under
Analysis')) and isnull(launch_datetime_scheduled, missions.launch_datetime_net) <
getdate()
    order by isnull(missions.launch_datetime_scheduled, launch_datetime_net)
desc
) as most_recent_mission_id on mission_rocket_stages.mission_id =
most_recent_mission_id.mission_id
order by stage_serial asc
```



Results | Messages

	stage_serial
1	EXC23-11003
2	EXC23-22001

11.5. Tárolt eljárás

11.5.1. Egy küldetés teljes reportja

A küldetés neve, státusza, leírása, legkorábbi indítási dátuma, [rakományok neve, típusa, tömege, gyártója, készültségi státusza, hozzárendelt pályák neve,

rakományokhoz tartozó szerződések címe, értéke], vagy ha emberes: [űrhajósok neve, életkoruk a küldetés indításakor, nemzetiségük, ügynökségük, szerződéseik címe, értéke]

```
use [Avantvia Mission Control Database]
go


drop procedure if exists cargo_mission_report print 'cargo_mission_report
already in db, overwritten'
drop procedure if exists crewed_mission_report print 'crewed_mission_report
already in db, overwritten'
drop procedure if exists get_mission_report print 'get_mission_report already
in db, overwritten'
go


create procedure cargo_mission_report
    @mission_id smallint
as
begin
    select missions.mission_name, mission_status.status_name as
mission_status_name, missions.mission_description, missions.launch_datetime_net,
           payloads.payload_name, payloads.payload_type,
payloads.payload_mass_g,
           payload_manufacturer.partner_name as manufacturer_name,
           payload_status.status_name as payload_status_name,
           target_orbits.orbit_name,
           payload_customer.partner_name as customer_name,
           contracts.contract_title, contracts.contract_value_eur
    from missions
           inner join status_codes as mission_status on missions.mission_id =
@mission_id and missions.mission_status_code = mission_status.status_code
           inner join payload_mission_manifests on missions.mission_id =
payload_mission_manifests.mission_id
           inner join payloads on payload_mission_manifests.payload_id =
payloads.payload_id
           inner join status_codes as payload_status on
payloads.payload_status_code = payload_status.status_code
           inner join partner_organizations as payload_manufacturer on
payloads.manufacturer_id = payload_manufacturer.partner_id
           inner join partner_organizations as payload_customer on
payloads.customer_id = payload_customer.partner_id
           inner join target_orbits on payload_mission_manifests.target_orbit_id
= target_orbits.target_orbit_id
           inner join contract_manifest_links on
payload_mission_manifests.manifest_id = contract_manifest_links.manifest_id
```

```
            inner join contracts on contract_manifest_links.contract_id =
contracts.contract_id
   end


   go

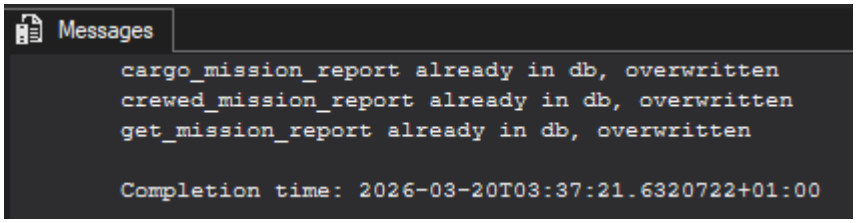

   create procedure crewed_mission_report
      @mission_id smallint
   as
   begin
      select missions.mission_name, mission_status.status_name as
mission_status_name, missions.mission_description, missions.launch_datetime_net,
            astronauts.first_name + ' ' + astronauts.last_name as full_name,
            datediff(year, date_of_birth, missions.launch_datetime_net)
            - iif(dateadd(year, datediff(year, date_of_birth,
cast(missions.launch_datetime_net as date)), date_of_birth)
            > cast(missions.launch_datetime_net as date), 1, 0) as
astronaut_age_at_launch,
            astronauts.nationality,
            astronaut_agency.partner_name as agency_name,
            isnull(agency_contracts.contract_title, 'Seat share') as
contract_title, agency_contracts.contract_value_eur
      from missions
            inner join status_codes as mission_status on
missions.mission_status_code = mission_status.status_code
            inner join astronaut_missions on missions.mission_id =
astronaut_missions.mission_id
            inner join astronauts on astronaut_missions.astronaut_id =
astronauts.astronaut_id
            inner join partner_organizations as astronaut_agency on
astronauts.agency_id = astronaut_agency.partner_id
            left join (
                  select distinct payload_mission_manifests.mission_id,
contracts.contract_id, contracts.contract_title, contracts.contract_value_eur,
contracts.contractor_id
            from contracts
                  inner join contract_manifest_links on contracts.contract_id =
contract_manifest_links.contract_id
                  inner join payload_mission_manifests on
contract_manifest_links.manifest_id = payload_mission_manifests.manifest_id
            ) as agency_contracts on missions.mission_id =
agency_contracts.mission_id and astronauts.agency_id =
agency_contracts.contractor_id
            where missions.mission_id = @mission_id
```

```
end

go

create procedure get_mission_report
    @mission_id smallint
as
begin
    declare @is_crewed bit
    select @is_crewed = is_crewed_mission
    from missions
    where mission_id = @mission_id

    if @is_crewed = 1 exec dbo.crewed_mission_report @mission_id
    else exec dbo.cargo_mission_report @mission_id
end
```

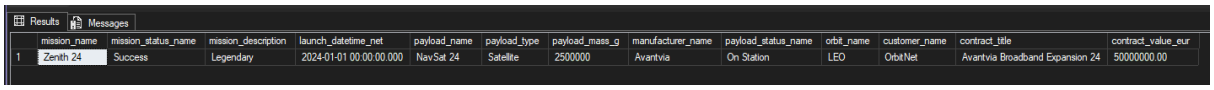


A teherszállító és emberes küldetéseket különválasztottam egy-egy tárolt eljárásba, így összesen 3 db tárolt eljárás kerül létrehozásra. A get_mission_report eljárást mindkét fajta küldetésre futtattam.

Teherszállító küldetés:

```
USE [Avantvia Mission Control Database]

EXEC [dbo].[get_mission_report] @mission_id = 24
```

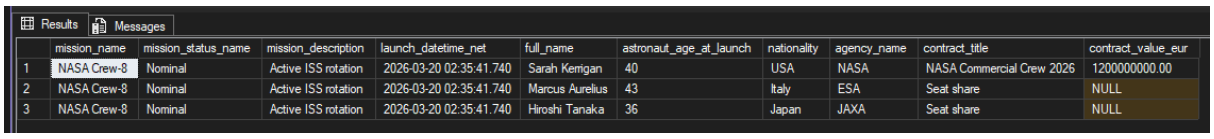


	mission_name	mission_status_name	mission_description	launch_datetime_net	payload_name	payload_type	payload_mass_g	manufacturer_name	payload_status_name	orbit_name	customer_name	contract_title	contract_value_eur
1	Zenith 24	Success	Legendary	2024-01-01 00:00:00.000	NavSat 24	Satellite	2500000	Avantvia	On Station	LEO	OrbitNet	Avantvia Broadband Expansion 24	50000000.00

Emberes küldetés:

```
USE [Avantvia Mission Control Database]

EXEC [dbo].[get_mission_report] @mission_id = 49
```

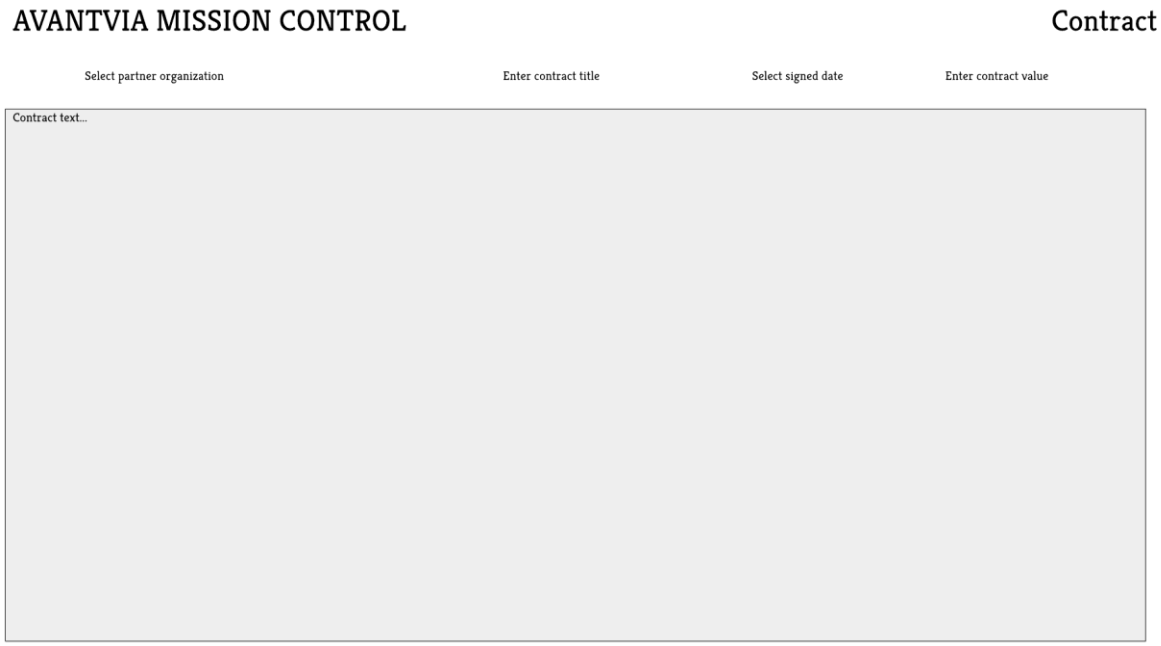


	mission_name	mission_status_name	mission_description	launch_datetime_net	full_name	astronaut_age_at_launch	nationality	agency_name	contract_title	contract_value_eur
1	NASA Crew-8	Nominal	Active ISS rotation	2026-03-20 02:35:41.740	Sarah Kerrigan	40	USA	NASA	NASA Commercial Crew 2026	1200000000.00
2	NASA Crew-8	Nominal	Active ISS rotation	2026-03-20 02:35:41.740	Marcus Aurelius	43	Italy	ESA	Seat share	NULL
3	NASA Crew-8	Nominal	Active ISS rotation	2026-03-20 02:35:41.740	Hiroshi Tanaka	36	Japan	JAXA	Seat share	NULL

Megjegyzés: az „ülésmegosztás” azt az esetet jelenti, amikor a NASA szerződik egy emberes repülés minden ülésére, és más űrügynökségek űrhajósait ültet bele.

Ezesetben a többi űrügynökségnek nincsen szerződése az Avantvia-val, csak a NASA-val. Az amerikai-orosz üléscsere program is így működik.

12. GUI tervek





Rocket stage

Enter stage type

Enter stage serial

Set status code

Next

13. Mellékletek

13.1. 1. számú melléklet: adatbázist létrehozó script

```
USE [master]
GO
CREATE DATABASE [Avantvia Mission Control Database]
 CONTAINMENT = NONE
 ON  PRIMARY
( NAME = N'Avantvia Mission Manager System', FILENAME = N'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\Avantvia Mission Manager System.mdf' , SIZE = 73728KB , MAXSIZE = UNLIMITED, FILEGROWTH = 65536KB )
 LOG ON
( NAME = N'Avantvia Mission Manager System_log', FILENAME = N'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\Avantvia Mission Manager System_log.ldf' , SIZE = 73728KB , MAXSIZE = 2048GB , FILEGROWTH = 65536KB )
 WITH CATALOG_COLLATION = DATABASE_DEFAULT, LEDGER = OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET COMPATIBILITY_LEVEL = 160
GO
IF (1 = FULLTEXTSERVICEPROPERTY('IsFullTextInstalled'))
begin
EXEC [Avantvia Mission Control Database].[dbo].[sp_fulltext_database] @action = 'enable'
end
GO
ALTER DATABASE [Avantvia Mission Control Database] SET ANSI_NULL_DEFAULT OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET ANSI_NULLS OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET ANSI_PADDING OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET ANSI_WARNINGS OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET ARITHABORT OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET AUTO_CLOSE OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET AUTO_SHRINK OFF
GO
```

```
   ALTER DATABASE [Avantvia Mission Control Database] SET AUTO_UPDATE_STATISTICS
ON
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET CURSOR_CLOSE_ON_COMMIT
OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET CURSOR_DEFAULT  GLOBAL
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET CONCAT_NULL_YIELDS_NULL
OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET NUMERIC_ROUNDABORT OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET QUOTED_IDENTIFIER OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET RECURSIVE_TRIGGERS OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET  DISABLE_BROKER
   GO
   ALTER     DATABASE     [Avantvia     Mission     Control     Database]     SET
AUTO_UPDATE_STATISTICS_ASYNC OFF
   GO
   ALTER     DATABASE     [Avantvia     Mission     Control     Database]     SET
DATE_CORRELATION_OPTIMIZATION OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET TRUSTWORTHY OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET ALLOW_SNAPSHOT_ISOLATION
OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET PARAMETERIZATION SIMPLE
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET READ_COMMITTED_SNAPSHOT
OFF
   GO
   ALTER  DATABASE [Avantvia Mission Control Database] SET  HONOR_BROKER_PRIORITY
OFF
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET RECOVERY SIMPLE
   GO
   ALTER DATABASE [Avantvia Mission Control Database] SET  MULTI_USER
   GO
```

```
ALTER DATABASE [Avantvia Mission Control Database] SET PAGE_VERIFY CHECKSUM
GO
ALTER DATABASE [Avantvia Mission Control Database] SET DB_CHAINING OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET FILESTREAM( NON_TRANSACTED_ACCESS = OFF )
GO
ALTER DATABASE [Avantvia Mission Control Database] SET TARGET_RECOVERY_TIME = 60 SECONDS
GO
ALTER DATABASE [Avantvia Mission Control Database] SET DELAYED_DURABILITY = DISABLED
GO
ALTER DATABASE [Avantvia Mission Control Database] SET ACCELERATED_DATABASE_RECOVERY = OFF
GO
ALTER DATABASE [Avantvia Mission Control Database] SET QUERY_STORE = ON
GO
ALTER DATABASE [Avantvia Mission Control Database] SET QUERY_STORE (OPERATION_MODE = READ_WRITE, CLEANUP_POLICY = (STALE_QUERY_THRESHOLD_DAYS = 30), DATA_FLUSH_INTERVAL_SECONDS = 900, INTERVAL_LENGTH_MINUTES = 60, MAX_STORAGE_SIZE_MB = 1000, QUERY_CAPTURE_MODE = AUTO, SIZE_BASED_CLEANUP_MODE = AUTO, MAX_PLANS_PER_QUERY = 200, WAIT_STATS_CAPTURE_MODE = ON)
GO
USE [Avantvia Mission Control Database]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [astronaut_missions](
    [astronaut_mission_id] [smallint] IDENTITY(1,1) NOT NULL,
    [mission_id] [smallint] NOT NULL,
    [astronaut_id] [smallint] NOT NULL,
    [astronaut_role] [varchar](50) NOT NULL,
    [is_backup] [bit] NOT NULL,
 CONSTRAINT [PK_astronaut_missions] PRIMARY KEY CLUSTERED
(
    [astronaut_mission_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
```

```
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [astronauts](
    [astronaut_id] [smallint] IDENTITY(1,1) NOT NULL,
    [first_name] [nvarchar](100) NOT NULL,
    [last_name] [nvarchar](100) NOT NULL,
    [date_of_birth] [date] NOT NULL,
    [nationality] [varchar](50) NOT NULL,
    [agency_id] [smallint] NOT NULL,
    [astronaut_status_code] [tinyint] NOT NULL,
 CONSTRAINT [PK_astronauts] PRIMARY KEY CLUSTERED
(
    [astronaut_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [contract_manifest_links](
    [manifest_id] [int] NOT NULL,
    [contract_id] [smallint] NOT NULL
) ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [contracts](
    [contract_id] [smallint] IDENTITY(1,1) NOT NULL,
    [contractor_id] [smallint] NOT NULL,
    [contract_title] [varchar](100) NOT NULL,
    [contract_signed_date] [date] NOT NULL,
    [contract_value_eur] [decimal](18, 2) NOT NULL,
```

```
    [contract_details] [nvarchar](max) NOT NULL,
 CONSTRAINT [PK_contracts] PRIMARY KEY CLUSTERED
(
    [contract_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [landing_targets](
    [landing_target_id] [smallint] IDENTITY(1,1) NOT NULL,
    [landing_target_site] [nvarchar](50) NULL,
    [landing_target_name] [varchar](25) NOT NULL,
    [landing_target_type] [varchar](15) NOT NULL,
    [landing_target_status_code] [tinyint] NOT NULL,
 CONSTRAINT [PK_landing_targets] PRIMARY KEY CLUSTERED
(
    [landing_target_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [launch_pads](
    [launch_pad_id] [smallint] IDENTITY(1,1) NOT NULL,
    [launch_site_id] [smallint] NOT NULL,
    [launch_pad_name] [varchar](25) NOT NULL,
    [launch_pad_longitude] [decimal](9, 6) NOT NULL,
    [launch_pad_latitude] [decimal](9, 6) NOT NULL,
    [launch_pad_status_code] [tinyint] NOT NULL,
 CONSTRAINT [PK_launch_pads] PRIMARY KEY CLUSTERED
(
    [launch_pad_id] ASC
```

```
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
   CREATE TABLE [launch_sites](
      [launch_site_id] [smallint] IDENTITY(1,1) NOT NULL,
      [launch_site_name] [nvarchar](25) NOT NULL,
      [launch_site_country] [varchar](25) NOT NULL,
    CONSTRAINT [PK_launch_sites] PRIMARY KEY CLUSTERED
   (
      [launch_site_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
   CREATE TABLE [mission_rocket_stages](
      [mission_stage_id] [smallint] IDENTITY(1,1) NOT NULL,
      [mission_id] [smallint] NOT NULL,
      [stage_id] [smallint] NOT NULL,
      [stage_number] [tinyint] NOT NULL,
      [stage_role] [varchar](25) NOT NULL,
    CONSTRAINT [PK_mission_rocket_stages] PRIMARY KEY CLUSTERED
   (
      [mission_stage_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
```

```
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [missions](
   [mission_id] [smallint] IDENTITY(1,1) NOT NULL,
   [mission_name] [nvarchar](50) NOT NULL,
   [launch_pad_id] [smallint] NOT NULL,
   [is_crewed_mission] [bit] NOT NULL,
   [launch_datetime_net] [datetime] NOT NULL,
   [launch_window_seconds] [int] NULL,
   [launch_datetime_scheduled] [datetime] NULL,
   [mission_status_code] [tinyint] NOT NULL,
   [mission_description] [nvarchar](max) NOT NULL,
   [post_mission_summary] [nvarchar](max) NULL,
 CONSTRAINT [PK_missions] PRIMARY KEY CLUSTERED
(
   [mission_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [partner_organizations](
   [partner_id] [smallint] IDENTITY(1,1) NOT NULL,
   [partner_name] [nvarchar](50) NOT NULL,
   [partner_type] [varchar](25) NOT NULL,
   [partner_country_of_origin] [varchar](25) NOT NULL,
   [partner_notes] [nvarchar](max) NOT NULL,
 CONSTRAINT [PK_partner_organizations] PRIMARY KEY CLUSTERED
(
   [partner_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY] TEXTIMAGE_ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
```

```
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [payload_mission_manifests](
    [manifest_id] [int] IDENTITY(1,1) NOT NULL,
    [mission_id] [smallint] NOT NULL,
    [payload_id] [int] NOT NULL,
    [target_orbit_id] [int] NOT NULL,
    [deploy_status_code] [tinyint] NOT NULL,
 CONSTRAINT [PK_payload_mission_manifests] PRIMARY KEY CLUSTERED
(
    [manifest_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
CREATE TABLE [payloads](
    [payload_id] [int] IDENTITY(1,1) NOT NULL,
    [payload_serial] [varchar](50) NOT NULL,
    [payload_name] [varchar](50) NOT NULL,
    [payload_type] [varchar](25) NOT NULL,
    [payload_mass_g] [int] NOT NULL,
    [manufacturer_id] [smallint] NOT NULL,
    [customer_id] [smallint] NOT NULL,
    [payload_status_code] [tinyint] NOT NULL,
 CONSTRAINT [PK_payloads] PRIMARY KEY CLUSTERED
(
    [payload_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
) ON [PRIMARY]
GO
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
```

```
   CREATE TABLE [rocket_stages](
      [stage_id] [smallint] IDENTITY(1,1) NOT NULL,
      [stage_type] [varchar](50) NOT NULL,
      [stage_serial] [varchar](50) NOT NULL,
      [stage_status_code] [tinyint] NOT NULL,
    CONSTRAINT [PK_rocket_stages] PRIMARY KEY CLUSTERED
   (
      [stage_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
   CREATE TABLE [stage_landing_target_links](
      [mission_stage_id] [smallint] NOT NULL,
      [landing_target_id] [smallint] NOT NULL
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
   CREATE TABLE [status_categories](
      [category_code] [tinyint] NOT NULL,
      [category_name] [varchar](25) NOT NULL,
    CONSTRAINT [PK_category_codes] PRIMARY KEY CLUSTERED
   (
      [category_code] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
```

```
   CREATE TABLE [status_category_links](
      [status_code] [tinyint] NOT NULL,
      [category_code] [tinyint] NOT NULL
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
   CREATE TABLE [status_codes](
      [status_code] [tinyint] NOT NULL,
      [status_name] [varchar](25) NOT NULL,
    CONSTRAINT [PK_status_codes] PRIMARY KEY CLUSTERED
   (
      [status_code] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
   SET ANSI_NULLS ON
   GO
   SET QUOTED_IDENTIFIER ON
   GO
   CREATE TABLE [target_orbits](
      [target_orbit_id] [int] IDENTITY(1,1) NOT NULL,
      [orbit_name] [varchar](25) NOT NULL,
      [apoapsis_cm] [bigint] NOT NULL,
      [periapsis_cm] [bigint] NOT NULL,
      [inclination_deg] [decimal](6, 3) NOT NULL,
      [is_interplanetary] [bit] NOT NULL,
      [target_body] [varchar](25) NULL,
      [orbit_notes] [varchar](4000) NOT NULL,
    CONSTRAINT [PK_target_orbits] PRIMARY KEY CLUSTERED
   (
      [target_orbit_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY = OFF,
ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON
[PRIMARY]
   ) ON [PRIMARY]
   GO
```

```
CREATE NONCLUSTERED INDEX [IX_astronaut_missionid] ON [astronaut_missions]
(
    [mission_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_astronaut_missions] ON [astronaut_missions]
(
    [astronaut_mission_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_astronaut_missions_astronaut] ON
[astronaut_missions]
(
    [astronaut_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_astronauts] ON [astronauts]
(
    [astronaut_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
SET ANSI_PADDING ON
GO
CREATE NONCLUSTERED INDEX [IX_astronauts_1] ON [astronauts]
(
    [last_name] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_astronauts_2] ON [astronauts]
(
    [agency_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
```

```
GO

CREATE NONCLUSTERED INDEX [IX_astronauts_3] ON [astronauts]

(

    [astronaut_status_code] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_contract_manifest_links] ON
[contract_manifest_links]

(

    [manifest_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_contract_manifest_links_1] ON
[contract_manifest_links]

(

    [contract_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_contracts] ON [contracts]

(

    [contract_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_contracts_1] ON [contracts]

(

    [contractor_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

SET ANSI_PADDING ON

GO

CREATE NONCLUSTERED INDEX [IX_contracts_2] ON [contracts]

(

    [contract_title] ASC
```

```
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_landing_targets] ON [landing_targets]

   (

      [landing_target_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   SET ANSI_PADDING ON

   GO

   CREATE NONCLUSTERED INDEX [IX_landing_targets_1] ON [landing_targets]

   (

      [landing_target_name] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_landing_targets_2] ON [landing_targets]

   (

      [landing_target_status_code] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_launch_pads] ON [launch_pads]

   (

      [launch_pad_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_launch_pads_1] ON [launch_pads]

   (

      [launch_site_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   SET ANSI_PADDING ON

   GO
```

```
CREATE NONCLUSTERED INDEX [IX_launch_pads_2] ON [launch_pads]
(
    [launch_pad_name] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_launch_pads_3] ON [launch_pads]
(
    [launch_pad_status_code] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_launch_sites] ON [launch_sites]
(
    [launch_site_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
SET ANSI_PADDING ON
GO
CREATE NONCLUSTERED INDEX [IX_launch_sites_1] ON [launch_sites]
(
    [launch_site_name] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
SET ANSI_PADDING ON
GO
CREATE NONCLUSTERED INDEX [IX_launch_sites_2] ON [launch_sites]
(
    [launch_site_country] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_mission_rocket_stages] ON [mission_rocket_stages]
(
    [mission_stage_id] ASC
```

```
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE      NONCLUSTERED      INDEX      [IX_mission_rocket_stages_1]      ON
[mission_rocket_stages]

   (

      [mission_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE      NONCLUSTERED      INDEX      [IX_mission_rocket_stages_2]      ON
[mission_rocket_stages]

   (

      [stage_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_missions] ON [missions]

   (

      [mission_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   SET ANSI_PADDING ON

   GO

   CREATE NONCLUSTERED INDEX [IX_missions_1] ON [missions]

   (

      [mission_name] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_missions_2] ON [missions]

   (

      [launch_pad_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_missions_3] ON [missions]
```

```
   (
      [is_crewed_mission] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE NONCLUSTERED INDEX [IX_missions_4] ON [missions]
   (
      [launch_datetime_net] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE NONCLUSTERED INDEX [IX_missions_5] ON [missions]
   (
      [mission_status_code] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE NONCLUSTERED INDEX [IX_partner_organizations] ON [partner_organizations]
   (
      [partner_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   SET ANSI_PADDING ON
   GO
   CREATE      NONCLUSTERED      INDEX      [IX_partner_organizations_1]      ON
[partner_organizations]
   (
      [partner_name] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE      NONCLUSTERED      INDEX      [IX_payload_mission_manifests]      ON
[payload_mission_manifests]
   (
      [payload_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
```

```
GO

CREATE NONCLUSTERED INDEX [IX_payload_mission_manifests_1] ON
[payload_mission_manifests]

(

    [target_orbit_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_payload_mission_manifests_2] ON
[payload_mission_manifests]

(

    [deploy_status_code] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_payloads] ON [payloads]

(

    [payload_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

SET ANSI_PADDING ON

GO

CREATE NONCLUSTERED INDEX [IX_payloads_1] ON [payloads]

(

    [payload_name] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_payloads_2] ON [payloads]

(

    [manufacturer_id] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

GO

CREATE NONCLUSTERED INDEX [IX_payloads_3] ON [payloads]

(

    [manufacturer_id] ASC
```

```
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE NONCLUSTERED INDEX [IX_payloads_4] ON [payloads]
   (
      [customer_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE NONCLUSTERED INDEX [IX_payloads_5] ON [payloads]
   (
      [payload_status_code] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   CREATE NONCLUSTERED INDEX [IX_rocket_stages] ON [rocket_stages]
   (
      [stage_id] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   SET ANSI_PADDING ON
   GO
   CREATE NONCLUSTERED INDEX [IX_rocket_stages_1] ON [rocket_stages]
   (
      [stage_type] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
   SET ANSI_PADDING ON
   GO
   CREATE NONCLUSTERED INDEX [IX_rocket_stages_2] ON [rocket_stages]
   (
      [stage_serial] ASC
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
   GO
```

```
CREATE NONCLUSTERED INDEX [IX_rocket_stages_3] ON [rocket_stages]
(
    [stage_status_code] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_stage_landing_target_links] ON
[stage_landing_target_links]
(
    [mission_stage_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_stage_landing_target_links_1] ON
[stage_landing_target_links]
(
    [landing_target_id] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_status_categories] ON [status_categories]
(
    [category_code] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
SET ANSI_PADDING ON
GO
CREATE NONCLUSTERED INDEX [IX_status_categories_1] ON [status_categories]
(
    [category_name] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
GO
CREATE NONCLUSTERED INDEX [IX_status_category_links] ON [status_category_links]
(
    [status_code] ASC
```

```
   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE      NONCLUSTERED      INDEX      [IX_status_category_links_1]      ON
[status_category_links]

   (

      [category_code] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_status_codes] ON [status_codes]

   (

      [status_code] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   SET ANSI_PADDING ON

   GO

   CREATE NONCLUSTERED INDEX [IX_status_codes_1] ON [status_codes]

   (

      [status_name] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   CREATE NONCLUSTERED INDEX [IX_target_orbits] ON [target_orbits]

   (

      [target_orbit_id] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO

   SET ANSI_PADDING ON

   GO

   CREATE NONCLUSTERED INDEX [IX_target_orbits_1] ON [target_orbits]

   (

      [orbit_name] ASC

   )WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, SORT_IN_TEMPDB = OFF,
DROP_EXISTING = OFF, ONLINE = OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

   GO
```

```
   ALTER TABLE [missions] ADD  CONSTRAINT [DF_missions_is_crewed_mission]  DEFAULT
((0)) FOR [is_crewed_mission]

   GO

   ALTER  TABLE  [missions]  ADD   CONSTRAINT  [DF_missions_launch_window_seconds]
DEFAULT ((0)) FOR [launch_window_seconds]

   GO

   ALTER        TABLE        [target_orbits]        ADD                CONSTRAINT
[DF_target_orbits_is_interplanetary]  DEFAULT ((0)) FOR [is_interplanetary]

   GO

   ALTER   TABLE   [astronaut_missions]      WITH   CHECK   ADD      CONSTRAINT
[FK_astronaut_missions_astronauts] FOREIGN KEY([astronaut_id])

   REFERENCES [astronauts] ([astronaut_id])

   GO

   ALTER        TABLE        [astronaut_missions]        CHECK        CONSTRAINT
[FK_astronaut_missions_astronauts]

   GO

   ALTER   TABLE   [astronaut_missions]      WITH   CHECK   ADD      CONSTRAINT
[FK_astronaut_missions_missions] FOREIGN KEY([mission_id])

   REFERENCES [missions] ([mission_id])

   GO

   ALTER        TABLE        [astronaut_missions]        CHECK        CONSTRAINT
[FK_astronaut_missions_missions]

   GO

   ALTER    TABLE    [astronauts]        WITH    CHECK    ADD        CONSTRAINT
[FK_astronauts_partner_organizations] FOREIGN KEY([agency_id])

   REFERENCES [partner_organizations] ([partner_id])

   GO

   ALTER TABLE [astronauts] CHECK CONSTRAINT [FK_astronauts_partner_organizations]

   GO

   ALTER    TABLE    [astronauts]        WITH    CHECK    ADD        CONSTRAINT
[FK_astronauts_status_codes] FOREIGN KEY([astronaut_status_code])

   REFERENCES [status_codes] ([status_code])

   GO

   ALTER TABLE [astronauts] CHECK CONSTRAINT [FK_astronauts_status_codes]

   GO

   ALTER   TABLE   [contract_manifest_links]     WITH   CHECK   ADD      CONSTRAINT
[FK_contract_manifest_links_contracts] FOREIGN KEY([contract_id])

   REFERENCES [contracts] ([contract_id])

   GO

   ALTER       TABLE       [contract_manifest_links]       CHECK       CONSTRAINT
[FK_contract_manifest_links_contracts]

   GO
```

```
   ALTER TABLE [contract_manifest_links] WITH CHECK ADD CONSTRAINT
[FK_contract_manifest_links_payload_mission_manifests] FOREIGN KEY([manifest_id])
   REFERENCES [payload_mission_manifests] ([manifest_id])
   GO
   ALTER TABLE [contract_manifest_links] CHECK CONSTRAINT
[FK_contract_manifest_links_payload_mission_manifests]
   GO
   ALTER TABLE [contracts] WITH CHECK ADD CONSTRAINT
[FK_contracts_partner_organizations] FOREIGN KEY([contractor_id])
   REFERENCES [partner_organizations] ([partner_id])
   GO
   ALTER TABLE [contracts] CHECK CONSTRAINT [FK_contracts_partner_organizations]
   GO
   ALTER TABLE [landing_targets] WITH CHECK ADD CONSTRAINT
[FK_landing_targets_status_codes] FOREIGN KEY([landing_target_status_code])
   REFERENCES [status_codes] ([status_code])
   GO
   ALTER TABLE [landing_targets] CHECK CONSTRAINT
[FK_landing_targets_status_codes]
   GO
   ALTER TABLE [launch_pads] WITH CHECK ADD CONSTRAINT
[FK_launch_pads_launch_sites] FOREIGN KEY([launch_site_id])
   REFERENCES [launch_sites] ([launch_site_id])
   GO
   ALTER TABLE [launch_pads] CHECK CONSTRAINT [FK_launch_pads_launch_sites]
   GO
   ALTER TABLE [launch_pads] WITH CHECK ADD CONSTRAINT
[FK_launch_pads_status_codes] FOREIGN KEY([launch_pad_status_code])
   REFERENCES [status_codes] ([status_code])
   GO
   ALTER TABLE [launch_pads] CHECK CONSTRAINT [FK_launch_pads_status_codes]
   GO
   ALTER TABLE [mission_rocket_stages] WITH CHECK ADD CONSTRAINT
[FK_mission_rocket_stages_missions] FOREIGN KEY([mission_id])
   REFERENCES [missions] ([mission_id])
   GO
   ALTER TABLE [mission_rocket_stages] CHECK CONSTRAINT
[FK_mission_rocket_stages_missions]
   GO
   ALTER TABLE [mission_rocket_stages] WITH CHECK ADD CONSTRAINT
[FK_mission_rocket_stages_rocket_stages] FOREIGN KEY([stage_id])
   REFERENCES [rocket_stages] ([stage_id])
   GO
```

```
   ALTER TABLE [mission_rocket_stages] CHECK CONSTRAINT
[FK_mission_rocket_stages_rocket_stages]

   GO

   ALTER TABLE [missions]  WITH CHECK ADD  CONSTRAINT [FK_missions_launch_pads]
FOREIGN KEY([launch_pad_id])

   REFERENCES [launch_pads] ([launch_pad_id])

   GO

   ALTER TABLE [missions] CHECK CONSTRAINT [FK_missions_launch_pads]

   GO

   ALTER TABLE [missions]  WITH CHECK ADD  CONSTRAINT [FK_missions_status_codes]
FOREIGN KEY([mission_status_code])

   REFERENCES [status_codes] ([status_code])

   GO

   ALTER TABLE [missions] CHECK CONSTRAINT [FK_missions_status_codes]

   GO

   ALTER TABLE [payload_mission_manifests]  WITH CHECK ADD  CONSTRAINT
[FK_payload_mission_manifests_missions] FOREIGN KEY([mission_id])

   REFERENCES [missions] ([mission_id])

   GO

   ALTER TABLE [payload_mission_manifests] CHECK CONSTRAINT
[FK_payload_mission_manifests_missions]

   GO

   ALTER TABLE [payload_mission_manifests]  WITH CHECK ADD  CONSTRAINT
[FK_payload_mission_manifests_payloads] FOREIGN KEY([payload_id])

   REFERENCES [payloads] ([payload_id])

   GO

   ALTER TABLE [payload_mission_manifests] CHECK CONSTRAINT
[FK_payload_mission_manifests_payloads]

   GO

   ALTER TABLE [payload_mission_manifests]  WITH CHECK ADD  CONSTRAINT
[FK_payload_mission_manifests_status_codes] FOREIGN KEY([deploy_status_code])

   REFERENCES [status_codes] ([status_code])

   GO

   ALTER TABLE [payload_mission_manifests] CHECK CONSTRAINT
[FK_payload_mission_manifests_status_codes]

   GO

   ALTER TABLE [payload_mission_manifests]  WITH CHECK ADD  CONSTRAINT
[FK_payload_mission_manifests_target_orbits] FOREIGN KEY([target_orbit_id])

   REFERENCES [target_orbits] ([target_orbit_id])

   GO

   ALTER TABLE [payload_mission_manifests] CHECK CONSTRAINT
[FK_payload_mission_manifests_target_orbits]

   GO
```

```
   ALTER     TABLE     [payloads]          WITH     CHECK     ADD          CONSTRAINT
[FK_payloads_partner_organizations] FOREIGN KEY([manufacturer_id])
   REFERENCES [partner_organizations] ([partner_id])
   GO
   ALTER TABLE [payloads] CHECK CONSTRAINT [FK_payloads_partner_organizations]
   GO
   ALTER     TABLE     [payloads]          WITH     CHECK     ADD          CONSTRAINT
[FK_payloads_partner_organizations1] FOREIGN KEY([customer_id])
   REFERENCES [partner_organizations] ([partner_id])
   GO
   ALTER TABLE [payloads] CHECK CONSTRAINT [FK_payloads_partner_organizations1]
   GO
   ALTER TABLE [payloads]  WITH CHECK ADD  CONSTRAINT [FK_payloads_status_codes]
FOREIGN KEY([payload_status_code])
   REFERENCES [status_codes] ([status_code])
   GO
   ALTER TABLE [payloads] CHECK CONSTRAINT [FK_payloads_status_codes]
   GO
   ALTER    TABLE    [rocket_stages]        WITH    CHECK    ADD        CONSTRAINT
[FK_rocket_stages_status_codes] FOREIGN KEY([stage_status_code])
   REFERENCES [status_codes] ([status_code])
   GO
   ALTER TABLE [rocket_stages] CHECK CONSTRAINT [FK_rocket_stages_status_codes]
   GO
   ALTER  TABLE  [stage_landing_target_links]    WITH  CHECK  ADD    CONSTRAINT
[FK_stage_landing_target_links_landing_targets] FOREIGN KEY([landing_target_id])
   REFERENCES [landing_targets] ([landing_target_id])
   GO
   ALTER      TABLE      [stage_landing_target_links]      CHECK      CONSTRAINT
[FK_stage_landing_target_links_landing_targets]
   GO
   ALTER  TABLE  [stage_landing_target_links]    WITH  CHECK  ADD    CONSTRAINT
[FK_stage_landing_target_links_mission_rocket_stages]                    FOREIGN
KEY([mission_stage_id])
   REFERENCES [mission_rocket_stages] ([mission_stage_id])
   GO
   ALTER      TABLE      [stage_landing_target_links]      CHECK      CONSTRAINT
[FK_stage_landing_target_links_mission_rocket_stages]
   GO
   ALTER   TABLE   [status_category_links]     WITH   CHECK   ADD     CONSTRAINT
[FK_status_category_links_status_categories] FOREIGN KEY([category_code])
   REFERENCES [status_categories] ([category_code])
   GO
```

```
   ALTER TABLE [status_category_links] CHECK CONSTRAINT
[FK_status_category_links_status_categories]
   GO
   ALTER TABLE [status_category_links] WITH CHECK ADD CONSTRAINT
[FK_status_category_links_status_codes] FOREIGN KEY([status_code])
   REFERENCES [status_codes] ([status_code])
   GO
   ALTER TABLE [status_category_links] CHECK CONSTRAINT
[FK_status_category_links_status_codes]
   GO
   ALTER TABLE [astronauts] WITH CHECK ADD CONSTRAINT [CK_astronauts_age] CHECK
((datediff(year,[date_of_birth],getdate())>=(21)))
   GO
   ALTER TABLE [astronauts] CHECK CONSTRAINT [CK_astronauts_age]
   GO
   ALTER TABLE [contracts] WITH CHECK ADD CONSTRAINT [CK_contracts_date] CHECK
(([contract_signed_date]>dateadd(month,(-1),getdate())))
   GO
   ALTER TABLE [contracts] CHECK CONSTRAINT [CK_contracts_date]
   GO
   ALTER TABLE [contracts] WITH CHECK ADD CONSTRAINT [CK_contracts_value] CHECK
(([contract_value_eur]>=(0)))
   GO
   ALTER TABLE [contracts] CHECK CONSTRAINT [CK_contracts_value]
   GO
   ALTER TABLE [launch_pads] WITH CHECK ADD CONSTRAINT [CK_launch_pads_lat]
CHECK (([launch_pad_latitude]>=(-90) AND [launch_pad_latitude]<=(90)))
   GO
   ALTER TABLE [launch_pads] CHECK CONSTRAINT [CK_launch_pads_lat]
   GO
   ALTER TABLE [launch_pads] WITH CHECK ADD CONSTRAINT [CK_launch_pads_long]
CHECK (([launch_pad_longitude]>=(-180) AND [launch_pad_longitude]<=(180)))
   GO
   ALTER TABLE [launch_pads] CHECK CONSTRAINT [CK_launch_pads_long]
   GO
   ALTER TABLE [mission_rocket_stages] WITH CHECK ADD CONSTRAINT
[CK_mission_rocket_stages_stageno] CHECK (([stage_number]>=(1) AND
[stage_number]<=(9)))
   GO
   ALTER TABLE [mission_rocket_stages] CHECK CONSTRAINT
[CK_mission_rocket_stages_stageno]
   GO
```

```
   ALTER TABLE [missions]  WITH CHECK ADD  CONSTRAINT [CK_missions_launchwindow]
CHECK  (([launch_window_seconds] IS NULL OR [launch_window_seconds]>=(0)))

   GO

   ALTER TABLE [missions] CHECK CONSTRAINT [CK_missions_launchwindow]

   GO

   ALTER TABLE [missions]  WITH CHECK ADD  CONSTRAINT [CK_missions_netdate] CHECK
(([launch_datetime_net]>getdate()))

   GO

   ALTER TABLE [missions] CHECK CONSTRAINT [CK_missions_netdate]

   GO

   ALTER TABLE [missions]  WITH CHECK ADD  CONSTRAINT [CK_missions_scheddate]
CHECK               (([launch_datetime_scheduled]       IS       NULL       OR
[launch_datetime_scheduled]>getdate()))

   GO

   ALTER TABLE [missions] CHECK CONSTRAINT [CK_missions_scheddate]

   GO

   ALTER TABLE [payloads]  WITH CHECK ADD  CONSTRAINT [CK_payloads_mass] CHECK
(([payload_mass_g]>(0)))

   GO

   ALTER TABLE [payloads] CHECK CONSTRAINT [CK_payloads_mass]

   GO

   ALTER TABLE [target_orbits]  WITH CHECK ADD  CONSTRAINT [CK_target_orbits_apo]
CHECK  (([apoapsis_cm]>=[periapsis_cm]))

   GO

   ALTER TABLE [target_orbits] CHECK CONSTRAINT [CK_target_orbits_apo]

   GO

   ALTER TABLE [target_orbits]  WITH CHECK ADD  CONSTRAINT [CK_target_orbits_incl]
CHECK  (([inclination_deg]>=(0) AND [inclination_deg]<=(180)))

   GO

   ALTER TABLE [target_orbits] CHECK CONSTRAINT [CK_target_orbits_incl]

   GO

   ALTER    TABLE    [target_orbits]       WITH    CHECK    ADD       CONSTRAINT
[CK_target_orbits_interplanetary]    CHECK       (([is_interplanetary]=(1)    AND
[target_body] IS NOT NULL OR [is_interplanetary]=(0) AND [target_body] IS NULL OR
[is_interplanetary]=(0) AND [target_body]='Earth'))

   GO

   ALTER TABLE [target_orbits] CHECK CONSTRAINT [CK_target_orbits_interplanetary]

   GO

   ALTER TABLE [target_orbits]  WITH CHECK ADD  CONSTRAINT [CK_target_orbits_peri]
CHECK  (([apoapsis_cm]>=[periapsis_cm]))

   GO

   ALTER TABLE [target_orbits] CHECK CONSTRAINT [CK_target_orbits_peri]

   GO
```

```
USE [master]
GO
ALTER DATABASE [Avantvia Mission Control Database] SET  READ_WRITE
GO
```

13.2. 2. számú melléklet: adatbázist adatokkal feltöltő script

Az adatok feltöltését végző script a Google gemini-3.1-pro-preview modellje segítségével készült.

```
USE [Avantvia Mission Control Database];
GO

EXEC sp_MSForEachTable 'ALTER TABLE ? NOCHECK CONSTRAINT ALL';
GO

DELETE FROM astronaut_missions;
DELETE FROM contract_manifest_links;
DELETE FROM payload_mission_manifests;
DELETE FROM stage_landing_target_links;
DELETE FROM mission_rocket_stages;
DELETE FROM missions;
DELETE FROM payloads;
DELETE FROM rocket_stages;
DELETE FROM contracts;
DELETE FROM astronauts;
DELETE FROM landing_targets;
DELETE FROM launch_pads;
DELETE FROM launch_sites;
DELETE FROM target_orbits;
DELETE FROM status_category_links;
DELETE FROM status_categories;
DELETE FROM status_codes;
DELETE FROM partner_organizations;
GO

DBCC CHECKIDENT ('astronaut_missions', RESEED, 0);
DBCC CHECKIDENT ('payload_mission_manifests', RESEED, 0);
DBCC CHECKIDENT ('missions', RESEED, 0);
```

```
DBCC CHECKIDENT ('payloads', RESEED, 0);
DBCC CHECKIDENT ('rocket_stages', RESEED, 0);
DBCC CHECKIDENT ('contracts', RESEED, 0);
DBCC CHECKIDENT ('astronauts', RESEED, 0);
DBCC CHECKIDENT ('landing_targets', RESEED, 0);
DBCC CHECKIDENT ('launch_pads', RESEED, 0);
DBCC CHECKIDENT ('launch_sites', RESEED, 0);
DBCC CHECKIDENT ('target_orbits', RESEED, 0);
DBCC CHECKIDENT ('partner_organizations', RESEED, 0);
GO


-- 1. Status Categories
INSERT INTO status_categories (category_code, category_name) VALUES
(1, 'Mission'), (2, 'Deployment'), (3, 'Rocket Stage'), (4, 'Landing Target'), (5, 'Launch Pad'), (6, 'Payload'), (7, 'Astronaut');


-- 2. Status Codes
INSERT INTO status_codes (status_code, status_name) VALUES
(1, 'Ready'), (2, 'Success'), (3, 'Failure'), (4, 'Partial Failure'), (5, 'Awaiting'), (6, 'On Hold'), (7, 'To Be Determined'), (8, 'Under Planning'), (9, 'Nominal'), (10, 'Degraded'), (11, 'Anomaly'), (12, 'Critical'), (13, 'Under Analysis'), (14, 'Cancelled'), (15, 'Propellant Loading'), (16, 'Weather Go'), (17, 'Weather Nogo'), (18, 'Weather Watch'), (19, 'Scrub'), (20, 'Signal Blackout'), (21, 'Signal Lost'), (22, 'Under Construction'), (23, 'Under Maintenance'), (24, 'On Route'), (25, 'On Mission'), (26, 'In Flight'), (27, 'Retired'), (28, 'Destroyed'), (29, 'Testing'), (30, 'Expended'), (31, 'Under Integration'), (32, 'Integrated'), (33, 'Occupied'), (34, 'Pad Hot'), (35, 'Pad Clear'), (36, 'Range Clear'), (37, 'On Station'), (38, 'Training'), (39, 'Recovering'), (40, 'Unavailable');


-- 3. Status Category Links
INSERT INTO status_category_links (category_code, status_code) VALUES
(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), (1, 9), (1, 10), (1, 11), (1, 12), (1, 13), (1, 14), (1, 15), (1, 16), (1, 17), (1, 18), (1, 19), (1, 20), (1, 21),
(2, 1), (2, 2), (2, 3), (2, 5), (2, 6), (2, 11),
(3, 22), (3, 23), (3, 24), (3, 1), (3, 25), (3, 26), (3, 27), (3, 28), (3, 29), (3, 9), (3, 10), (3, 11), (3, 12), (3, 30), (3, 31), (3, 32),
(4, 22), (4, 23), (4, 24), (4, 1), (4, 25), (4, 9), (4, 10), (4, 11), (4, 12), (4, 16), (4, 17), (4, 18),
(5, 22), (5, 23), (5, 1), (5, 33), (5, 11), (5, 34), (5, 35), (5, 36),
(6, 22), (6, 23), (6, 24), (6, 1), (6, 29), (6, 28), (6, 27), (6, 25), (6, 9), (6, 10), (6, 11), (6, 12), (6, 13), (6, 37), (6, 5),
(7, 27), (7, 38), (7, 25), (7, 39), (7, 11), (7, 40), (7, 1); -- Included 1 for Ready
```

```
-- 4. Partner Organizations
SET IDENTITY_INSERT partner_organizations ON;
INSERT INTO partner_organizations (partner_id, partner_name, partner_type, partner_country_of_origin, partner_notes) VALUES
(1, 'Avantvia', 'Company', 'USA', 'Internal'),
(2, 'NASA', 'Agency', 'USA', 'NASA'),
(3, 'ESA', 'Agency', 'Europe', 'ESA'),
(4, 'JAXA', 'Agency', 'Japan', 'JAXA'),
(5, 'OrbitNet', 'Company', 'USA', 'Sat Internet'),
(6, 'MIT', 'University', 'USA', 'MIT'),
(7, 'Stanford', 'University', 'USA', 'Stanford'),
(8, 'Oxford', 'University', 'UK', 'Oxford'),
(9, 'Caltech', 'University', 'USA', 'Caltech'),
(10, 'University of Tokyo', 'University', 'Japan', 'UTokyo'),
(11, 'Max Planck Institute', 'Institute', 'Germany', 'Max Planck');
SET IDENTITY_INSERT partner_organizations OFF;


-- 5. Launch Sites & Pads
SET IDENTITY_INSERT launch_sites ON;
INSERT INTO launch_sites (launch_site_id, launch_site_name, launch_site_country) VALUES (1, 'Cape Orizon', 'USA'), (2, 'Vandenberg SFB', 'USA'), (3, 'Guiana Space Centre', 'French Guiana');
SET IDENTITY_INSERT launch_sites OFF;


SET IDENTITY_INSERT launch_pads ON;
INSERT INTO launch_pads (launch_pad_id, launch_site_id, launch_pad_name, launch_pad_longitude, launch_pad_latitude, launch_pad_status_code) VALUES
(1, 1, 'Pad A', -80.5, 28.5, 1), (2, 1, 'Pad B', -80.6, 28.5, 1), (3, 1, 'Pad C', -80.5, 28.6, 1),
(4, 2, 'Pad D', -120.6, 34.6, 1), (5, 3, 'Pad E', -52.6, 5.2, 1);
SET IDENTITY_INSERT launch_pads OFF;


-- 6. Landing Targets
SET IDENTITY_INSERT landing_targets ON;
INSERT INTO landing_targets (landing_target_id, landing_target_name, landing_target_site, landing_target_type, landing_target_status_code) VALUES
(1, 'Ad Astra', 'Atlantic Ocean', 'ASDS', 1), (2, 'Per Aspera', 'Atlantic Ocean', 'ASDS', 1),
(3, 'Audentes', 'Pacific Ocean', 'ASDS', 1), (4, 'Ground Pad 1', 'Cape Orizon', 'Ground Pad', 1),
```

```
   (5, 'Ground Pad 2', 'Cape Orizon', 'Ground Pad', 1), (6, 'Ground Pad 3',
'Vandenberg SFB', 'Ground Pad', 1);

   SET IDENTITY_INSERT landing_targets OFF;


   -- 7. Target Orbits

   SET IDENTITY_INSERT target_orbits ON;

   INSERT INTO target_orbits (target_orbit_id, orbit_name, apoapsis_cm,
periapsis_cm, inclination_deg, is_interplanetary, target_body, orbit_notes) VALUES

   (1, 'LEO', 20000000, 20000000, 51.6, 0, 'Earth', 'LEO'),

   (2, 'ISS', 40000000, 40000000, 51.6, 0, 'Earth', 'ISS'),

   (3, 'SSO', 50000000, 50000000, 97.4, 0, 'Earth', 'SSO'),

   (4, 'GTO', 3500000000, 20000000, 28.5, 0, 'Earth', 'GTO'),

   (5, 'GEO', 3500000000, 3500000000, 0.0, 0, 'Earth', 'GEO'),

   (6, 'TLI', 38440000000, 20000000, 28.5, 1, 'Moon', 'Moon'),

   (7, 'MTO', 22500000000000, 20000000, 0.0, 1, 'Mars', 'Mars');

   SET IDENTITY_INSERT target_orbits OFF;


   -- 8. Astronauts

   SET IDENTITY_INSERT astronauts ON;

   INSERT INTO astronauts (astronaut_id, first_name, last_name, date_of_birth,
nationality, agency_id, astronaut_status_code) VALUES

   (1, 'Sarah', 'Kerrigan', '1985-05-12', 'USA', 2, 25),

   (2, 'Marcus', 'Aurelius', '1982-10-04', 'Italy', 3, 25),

   (3, 'Hiroshi', 'Tanaka', '1990-01-22', 'Japan', 4, 25),

   (4, 'Elena', 'Rostova', '1988-08-15', 'USA', 1, 25),

   (5, 'James', 'Holden', '1975-03-30', 'USA', 2, 1),

   (6, 'Naomi', 'Nagata', '1992-11-11', 'UK', 3, 1),

   (7, 'Amos', 'Burton', '1980-07-07', 'USA', 1, 1),

   (8, 'Alex', 'Kamal', '1984-02-14', 'USA', 2, 27),

   (9, 'Chrisjen', 'Avasarala', '1965-09-09', 'India', 1, 27),

   (10, 'Bobbie', 'Draper', '1989-12-25', 'New Zealand', 3, 39),

   (11, 'Joe', 'Miller', '1985-01-10', 'USA', 2, 1);

   SET IDENTITY_INSERT astronauts OFF;


   -- 9. Rocket Stages

   SET IDENTITY_INSERT rocket_stages ON;

   INSERT INTO rocket_stages (stage_id, stage_type, stage_serial,
stage_status_code) VALUES

   (1, 'Excelsior 2 Stage 1', 'EXC23-11001', 1), -- The Legend

   (2, 'Excelsior 2 Stage 1', 'EXC23-11002', 1),

   (3, 'Excelsior 2 Stage 1', 'EXC23-11003', 1),
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   (4, 'Excelsior 2 Stage 1', 'EXC23-11004', 1),
   (5, 'Excelsior 2 Stage 1', 'EXC23-11005', 1),
   (6, 'Excelsior 1 Stage 1', 'EXC11-10001', 27),
   (7, 'Excelsior 1 Stage 1', 'EXC11-10002', 28),
   (8, 'Excelsior 1 Stage 1', 'EXC11-10003', 30),
   (9, 'Excelsior 1 Stage 2', 'EXC11-20001', 30),
   (10, 'Excelsior Nova Stage 1', 'EXCN1-10001', 28),
   (11, 'Excelsior Nova Stage 2', 'EXCN1-20001', 28),
   (12, 'Excelsior Nova Stage 1', 'EXCN1-10002', 1),
   (13, 'Excelsior Nova Stage 2', 'EXCN1-20002', 1);
   SET IDENTITY_INSERT rocket_stages OFF;


   BEGIN TRANSACTION;
   BEGIN TRY
   DECLARE @MissionID INT, @PayloadID INT, @ManifestID INT, @ContractID INT,
@StageLinkID INT;


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Commercial Delivery 1',
'2020-01-01', 50000000.00, '...');
   SET @ContractID = SCOPE_IDENTITY();
   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net, launch_datetime_scheduled, mission_status_code,
mission_description, post_mission_summary)
   VALUES ('Orion 1', 2, 0, DATEADD(MONTH, 2, '2020-01-01'), DATEADD(MONTH, 2,
'2020-01-01'), 2, 'Legendary', 'Success');
   SET @MissionID = SCOPE_IDENTITY();
   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)
   VALUES ('LEG-1', 'EarthObs 1', 'Satellite', 2500000, 1, 5, 37);
   SET @PayloadID = SCOPE_IDENTITY();
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);
   SET @ManifestID = SCOPE_IDENTITY();
   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);
   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21001', 30);
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');
   SET @StageLinkID = SCOPE_IDENTITY();
```

```
   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 2);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Scientific 2', '2020-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Helios 2', 3, 0, DATEADD(MONTH, 4, '2020-01-01'), DATEADD(MONTH, 4,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-2', 'Recon 2', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21002', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 3);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Broadband Expansion 3',
'2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Nyx 3', 1, 0, DATEADD(MONTH, 6, '2020-01-01'), DATEADD(MONTH, 6,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-3', 'EarthObs 3', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21003', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 4);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Gov Ops 4', '2020-01-
01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Vanguard 4', 2, 0, DATEADD(MONTH, 8, '2020-01-01'), DATEADD(MONTH, 8,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-4', 'Weather 4', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21004', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 5);
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Scientific 5', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Zenith 5', 3, 0, DATEADD(MONTH, 10, '2020-01-01'), DATEADD(MONTH, 10, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-5', 'EarthObs 5', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-21005', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Broadband Expansion 6', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Odin 6', 1, 0, DATEADD(MONTH, 12, '2020-01-01'), DATEADD(MONTH, 12, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-6', 'Recon 6', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();
```

```
   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21006', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 2);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Global Network 7',
'2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Polaris 7', 2, 0, DATEADD(MONTH, 14, '2020-01-01'), DATEADD(MONTH, 14,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-7', 'EarthObs 7', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21007', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 3);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Global Network 8',
'2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();
```

```
   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Sirius 8', 3, 0, DATEADD(MONTH, 16, '2020-01-01'), DATEADD(MONTH, 16,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-8', 'NavSat 8', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21008', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 4);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Gov Ops 9', '2020-01-
01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Pioneer 9', 1, 0, DATEADD(MONTH, 18, '2020-01-01'), DATEADD(MONTH, 18,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-9', 'NavSat 9', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21009', 30);
```

```
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 5);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Gov Ops 10', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Sirius 10', 2, 0, DATEADD(MONTH, 20, '2020-01-01'), DATEADD(MONTH, 20, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-10', 'Recon 10', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-21010', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Commercial Delivery 11', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Zeus 11', 3, 0, DATEADD(MONTH, 22, '2020-01-01'), DATEADD(MONTH, 22, '2020-01-01'), 2, 'Legendary', 'Success');
```

```
   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-11', 'Recon 11', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21011', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 2);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Scientific 12', '2020-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Pioneer 12', 1, 0, DATEADD(MONTH, 24, '2020-01-01'), DATEADD(MONTH,
24, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-12', 'Weather 12', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21012', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');
```

```
   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 3);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Global Network 13',
'2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Zeus 13', 2, 0, DATEADD(MONTH, 26, '2020-01-01'), DATEADD(MONTH, 26,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-13', 'NavSat 13', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21013', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 4);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Scientific 14', '2020-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Apex 14', 3, 0, DATEADD(MONTH, 28, '2020-01-01'), DATEADD(MONTH, 28,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-14', 'Commsat 14', 'Satellite', 2500000, 1, 5, 37);
```

```
   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21014', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 5);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Broadband Expansion
15', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net, launch_datetime_scheduled, mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Apex 15', 1, 0, DATEADD(MONTH, 30, '2020-01-01'), DATEADD(MONTH, 30,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-15', 'Recon 15', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21015', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Gov Ops 16', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Chronos 16', 2, 0, DATEADD(MONTH, 32, '2020-01-01'), DATEADD(MONTH, 32, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-16', 'Weather 16', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-21016', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 2);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Broadband Expansion 17', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Pioneer 17', 3, 0, DATEADD(MONTH, 34, '2020-01-01'), DATEADD(MONTH, 34, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-17', 'Relay 17', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();
```

```
   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21017', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 3);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Scientific 18', '2020-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Titan 18', 1, 0, DATEADD(MONTH, 36, '2020-01-01'), DATEADD(MONTH, 36,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-18', 'Commsat 18', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21018', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 4);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Gov Ops 19', '2020-01-
01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();
```

```
   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Vanguard 19', 2, 0, DATEADD(MONTH, 38, '2020-01-01'), DATEADD(MONTH,
38, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-19', 'NavSat 19', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21019', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 5);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Global Network 20',
'2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Pioneer 20', 3, 0, DATEADD(MONTH, 40, '2020-01-01'), DATEADD(MONTH,
40, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-20', 'Weather 20', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21020', 30);
```

```
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Commercial Delivery 21', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Titan 21', 1, 0, DATEADD(MONTH, 42, '2020-01-01'), DATEADD(MONTH, 42, '2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-21', 'Weather 21', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-21021', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 2);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Avantvia Global Network 22', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Sirius 22', 2, 0, DATEADD(MONTH, 44, '2020-01-01'), DATEADD(MONTH, 44, '2020-01-01'), 2, 'Legendary', 'Success');
```

```
   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-22', 'EarthObs 22', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21022', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 3);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Scientific 23', '2020-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Zenith 23', 3, 0, DATEADD(MONTH, 46, '2020-01-01'), DATEADD(MONTH, 46,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-23', 'NavSat 23', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21023', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');
```

```
   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 4);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Broadband Expansion
24', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Zenith 24', 1, 0, DATEADD(MONTH, 48, '2020-01-01'), DATEADD(MONTH, 48,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-24', 'NavSat 24', 'Satellite', 2500000, 1, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21024', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 5);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Avantvia Commercial Delivery
25', '2020-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Odin 25', 2, 0, DATEADD(MONTH, 50, '2020-01-01'), DATEADD(MONTH, 50,
'2020-01-01'), 2, 'Legendary', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('LEG-25', 'Weather 25', 'Satellite', 2500000, 1, 5, 37);
```

```
   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-21025', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 1, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Commercial Delivery 1',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate1 DATETIME = DATEADD(DAY, -(1 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Apex Block B-1', CASE WHEN 1 % 5 = 0 THEN 4 ELSE (1 % 3) + 1 END, 0,
@CadenceDate1, @CadenceDate1, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-1', 'EarthObs 101', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22001', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 1 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Broadband Expansion 2', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate2 DATETIME = DATEADD(DAY, -(2 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Hermes Block B-2', CASE WHEN 2 % 5 = 0 THEN 4 ELSE (2 % 3) + 1 END, 0, @CadenceDate2, @CadenceDate2, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-2', 'Recon 102', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-22002', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, CASE WHEN 2 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Gov Ops 3', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate3 DATETIME = DATEADD(DAY, -(3 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Chronos Block B-3', CASE WHEN 3 % 5 = 0 THEN 4 ELSE (3 % 3) + 1 END, 0, @CadenceDate3, @CadenceDate3, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-3', 'NavSat 103', 'Satellite', 100000, 5, 5, 37);
```

```
   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22003', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 3 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Commercial Delivery 4',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate4 DATETIME = DATEADD(DAY, -(4 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Horizon Block B-4', CASE WHEN 4 % 5 = 0 THEN 4 ELSE (4 % 3) + 1 END,
0, @CadenceDate4, @CadenceDate4, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-4', 'Relay 104', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22004', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 4 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();
```

```
   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Commercial Delivery 5',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate5 DATETIME = DATEADD(DAY, -(5 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Helios Block B-5', CASE WHEN 5 % 5 = 0 THEN 4 ELSE (5 % 3) + 1 END, 0,
@CadenceDate5, @CadenceDate5, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-5', 'EarthObs 105', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22005', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 5 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Broadband Expansion 6',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate6 DATETIME = DATEADD(DAY, -(6 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Chronos Block B-6', CASE WHEN 6 % 5 = 0 THEN 4 ELSE (6 % 3) + 1 END,
0, @CadenceDate6, @CadenceDate6, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();
```

```
   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-6', 'EarthObs 106', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22006', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 6 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Gov Ops 7', '2024-01-01',
20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate7 DATETIME = DATEADD(DAY, -(7 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Horizon Block B-7', CASE WHEN 7 % 5 = 0 THEN 4 ELSE (7 % 3) + 1 END,
0, @CadenceDate7, @CadenceDate7, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-7', 'Recon 107', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22007', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, CASE WHEN 7 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Broadband Expansion 8', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate8 DATETIME = DATEADD(DAY, -(8 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Vega Block B-8', CASE WHEN 8 % 5 = 0 THEN 4 ELSE (8 % 3) + 1 END, 0, @CadenceDate8, @CadenceDate8, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-8', 'Weather 108', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-22008', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, CASE WHEN 8 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Scientific 9', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate9 DATETIME = DATEADD(DAY, -(9 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)
```

```
   VALUES ('Apollo Block B-9', CASE WHEN 9 % 5 = 0 THEN 4 ELSE (9 % 3) + 1 END, 0,
@CadenceDate9, @CadenceDate9, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-9', 'NavSat 109', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22009', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 9 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Global Network 10',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate10 DATETIME = DATEADD(DAY, -(10 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net, launch_datetime_scheduled, mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Titan Block B-10', CASE WHEN 10 % 5 = 0 THEN 4 ELSE (10 % 3) + 1 END,
0, @CadenceDate10, @CadenceDate10, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-10', 'NavSat 110', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22010', 30);
```

```
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 10 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Commercial Delivery 11',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate11 DATETIME = DATEADD(DAY, -(11 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Vanguard Block B-11', CASE WHEN 11 % 5 = 0 THEN 4 ELSE (11 % 3) + 1
END, 0, @CadenceDate11, @CadenceDate11, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-11', 'Commsat 111', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22011', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 11 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Global Network 12',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate12 DATETIME = DATEADD(DAY, -(12 * 15), GETDATE());
```

```
   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Nyx Block B-12', CASE WHEN 12 % 5 = 0 THEN 4 ELSE (12 % 3) + 1 END, 0,
@CadenceDate12, @CadenceDate12, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-12', 'NavSat 112', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22012', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 12 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Gov Ops 13', '2024-01-
01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate13 DATETIME = DATEADD(DAY, -(13 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Sirius Block B-13', CASE WHEN 13 % 5 = 0 THEN 4 ELSE (13 % 3) + 1 END,
0, @CadenceDate13, @CadenceDate13, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-13', 'NavSat 113', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);
```

```
   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22013', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 13 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Scientific 14', '2024-01-
01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate14 DATETIME = DATEADD(DAY, -(14 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Pioneer Block B-14', CASE WHEN 14 % 5 = 0 THEN 4 ELSE (14 % 3) + 1
END, 0, @CadenceDate14, @CadenceDate14, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-14', 'Recon 114', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22014', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 14 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Broadband Expansion 15',
'2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();
```

```
   DECLARE @CadenceDate15 DATETIME = DATEADD(DAY, -(15 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Pioneer Block B-15', CASE WHEN 15 % 5 = 0 THEN 4 ELSE (15 % 3) + 1 END, 0, @CadenceDate15, @CadenceDate15, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-15', 'NavSat 115', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-22015', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, CASE WHEN 15 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Commercial Delivery 16', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate16 DATETIME = DATEADD(DAY, -(16 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Apollo Block B-16', CASE WHEN 16 % 5 = 0 THEN 4 ELSE (16 % 3) + 1 END, 0, @CadenceDate16, @CadenceDate16, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-16', 'Recon 116', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();
```

```
   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22016', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 16 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Gov Ops 17', '2024-01-
01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate17 DATETIME = DATEADD(DAY, -(17 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Orion Block B-17', CASE WHEN 17 % 5 = 0 THEN 4 ELSE (17 % 3) + 1 END,
0, @CadenceDate17, @CadenceDate17, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-17', 'EarthObs 117', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22017', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 17 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Gov Ops 18', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate18 DATETIME = DATEADD(DAY, -(18 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Apex Block B-18', CASE WHEN 18 % 5 = 0 THEN 4 ELSE (18 % 3) + 1 END, 0, @CadenceDate18, @CadenceDate18, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-18', 'EarthObs 118', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-22018', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, CASE WHEN 18 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage 1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (5, 'Global Global Network 19', '2024-01-01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate19 DATETIME = DATEADD(DAY, -(19 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Zenith Block B-19', CASE WHEN 19 % 5 = 0 THEN 4 ELSE (19 % 3) + 1 END, 0, @CadenceDate19, @CadenceDate19, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-19', 'Weather 119', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22019', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 19 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (5, 'Global Gov Ops 20', '2024-01-
01', 20000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   DECLARE @CadenceDate20 DATETIME = DATEADD(DAY, -(20 * 15), GETDATE());

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Orion Block B-20', CASE WHEN 20 % 5 = 0 THEN 4 ELSE (20 % 3) + 1 END,
0, @CadenceDate20, @CadenceDate20, 2, 'Cadence', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CAD-20', 'Relay 120', 'Satellite', 100000, 5, 5, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-22020', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, CASE WHEN 20 % 2 = 0 THEN 2 ELSE 3 END, 1, 'Stage
1');

   SET @StageLinkID = SCOPE_IDENTITY();

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (@StageLinkID, 1);
```

```
-- NASA 1

INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (2, 'NASA Earth Sciences 2024', '2023-01-01', 400000000.00, '...');

SET @ContractID = SCOPE_IDENTITY();

INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

VALUES ('NASA Landsat 10', 1, 0, '2024-06-15 10:00:00', '2024-06-15 10:00:00', 2, 'Earth Obs', 'Success');

SET @MissionID = SCOPE_IDENTITY();

-- MFR is Avantvia, Customer is NASA

INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

VALUES ('L-10', 'Landsat 10', 'Satellite', 2000000, 1, 2, 37);

SET @PayloadID = SCOPE_IDENTITY();

INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 2);

SET @ManifestID = SCOPE_IDENTITY();

INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-23001', 30);

INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 2, 1, 'Stage 1');

INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (SCOPE_IDENTITY(), 4);


-- NASA 2

INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (2, 'NASA Deep Space Probes', '2023-05-01', 750000000.00, '...');

SET @ContractID = SCOPE_IDENTITY();

INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

VALUES ('NASA Deep Space 1', 2, 0, '2024-10-10 10:00:00', '2024-10-10 10:00:00', 2, 'Deep space', 'Success');

SET @MissionID = SCOPE_IDENTITY();

INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

VALUES ('DS-1', 'Deep Space Climate Obs', 'Satellite', 3000000, 1, 2, 37);
```

```
   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 6, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-23002', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 3, 1, 'Stage 1');

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (SCOPE_IDENTITY(), 1);


   -- NASA 3 (Crew-7)

   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (2, 'NASA CCtCap 2', '2023-08-01', 1200000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('NASA Crew-7', 1, 1, '2025-02-15 10:00:00', '2025-02-15 10:00:00', 2, 'Crew rot', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   -- Avantvia owns capsule (1), Avantvia manufactured it (1). NASA contract above covers launch

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('V-CR-7', 'Viator Crew Capsule 7', 'Capsule', 12500000, 1, 1, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 2, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 2 Stage 2', 'EXC23-23003', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 2, 1, 'Stage 1');

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id) VALUES (SCOPE_IDENTITY(), 4);
```

```
   INSERT INTO astronaut_missions (mission_id, astronaut_id, astronaut_role,
is_backup) VALUES (@MissionID, 5, 'Commander', 0), (@MissionID, 6, 'Pilot', 0);


   -- ONGOING CREW MISSION (Crew-8)

   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (2, 'NASA Commercial Crew 2026',
'2024-01-01', 1200000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   -- mission_status_code = 9 (Nominal)

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('NASA Crew-8', 1, 1, GETDATE(), GETDATE(), 9, 'Active ISS rotation',
NULL);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('V-CR-8', 'Viator Crew Capsule 8', 'Capsule', 12500000, 1, 1, 9);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 2, 9);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-24001', 26);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 3, 1, 'Stage 1');

   INSERT INTO astronaut_missions (mission_id, astronaut_id, astronaut_role,
is_backup) VALUES (@MissionID, 1, 'Commander', 0), (@MissionID, 2, 'Pilot', 0),
(@MissionID, 3, 'Mission Specialist', 0);


   -- Extra Crew Missions

   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (2, 'NASA CCtCap 1', '2018-09-16',
2600000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();


   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('NASA Crew-5', 1, 1, '2023-05-10 10:00:00', '2023-05-10 10:00:00', 2,
'Crew rot', 'Success');

   SET @MissionID = SCOPE_IDENTITY();
```

```
   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('V-CR-5', 'Viator Crew Capsule 5', 'Capsule', 12500000, 1, 1, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 2, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-24005', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 2, 1, 'Stage 1');

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (SCOPE_IDENTITY(), 4);

   INSERT INTO astronaut_missions (mission_id, astronaut_id, astronaut_role,
is_backup) VALUES (@MissionID, 5, 'Commander', 0), (@MissionID, 6, 'Pilot', 0);


   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('NASA Crew-6', 1, 1, '2024-03-01 10:00:00', '2024-03-01 10:00:00', 2,
'Crew rot', 'Success');

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('V-CR-6', 'Viator Crew Capsule 6', 'Capsule', 12500000, 1, 1, 37);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 2, 2);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-24006', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 2, 1, 'Stage 1');

   INSERT INTO stage_landing_target_links (mission_stage_id, landing_target_id)
VALUES (SCOPE_IDENTITY(), 4);

   INSERT INTO astronaut_missions (mission_id, astronaut_id, astronaut_role,
is_backup) VALUES (@MissionID, 1, 'Commander', 0), (@MissionID, 2, 'Pilot', 0);
```

```
   -- Cosmic-Share 7 (11 payloads)

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Cosmic-Share 7', 4, 0, '2026-11-15 14:00:00', '2026-11-15 14:00:00',
8, 'Mega Rideshare 1', NULL);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-25001', 1);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 2, 1, 'Stage 1');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((7), 'University Rideshare Nyx 1',
'2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-1', 'Student Nyx 1', 'CubeSat', 5000, 6, (7), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((8), 'University Rideshare Helios
2', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-2', 'Research Odin 2', 'CubeSat', 5000, 6, (8), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((9), 'University Rideshare Polaris 3', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-3', 'Tech Chronos 3', 'CubeSat', 5000, 6, (9), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((10), 'University Rideshare Nova 4', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-4', 'Academic Horizon 4', 'CubeSat', 5000, 6, (10), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((11), 'University Rideshare Ares 5', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-5', 'Student Zeus 5', 'CubeSat', 5000, 6, (11), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((6), 'University Rideshare Zeus 6', '2025-01-01', 500000.00, '...');
```

```
   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-6', 'Tech Sol 6', 'CubeSat', 5000, 6, (6), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((7), 'University Rideshare Hermes
7', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-7', 'Tech Horizon 7', 'CubeSat', 5000, 6, (7), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((8), 'University Rideshare Titan 8',
'2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-8', 'Student Vanguard 8', 'CubeSat', 5000, 6, (8), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((9), 'University Rideshare Luna 9',
'2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)
```

```
   VALUES ('CS7-9', 'Academic Apex 9', 'CubeSat', 5000, 6, (9), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((10), 'University Rideshare Titan
10', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-10', 'Research Polaris 10', 'CubeSat', 5000, 6, (10), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((11), 'University Rideshare Apex
11', '2025-01-01', 500000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS7-11', 'Research Vega 11', 'CubeSat', 5000, 6, (11), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   -- Cosmic-Share 8 (6 payloads)

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Cosmic-Share 8', 4, 0, '2027-02-15 14:00:00', '2027-02-15 14:00:00',
8, 'Mega Rideshare 2', NULL);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-25002', 1);
```

```
   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, 3, 1, 'Stage 1');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((8), 'University Rideshare Zeus 12', '2025-02-01', 600000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS8-1', 'Research Sirius 12', 'CubeSat', 8000, 7, (8), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((9), 'University Rideshare Chronos 13', '2025-02-01', 600000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS8-2', 'Academic Ares 13', 'CubeSat', 8000, 7, (9), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES ((10), 'University Rideshare Pioneer 14', '2025-02-01', 600000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS8-3', 'Tech Argus 14', 'CubeSat', 8000, 7, (10), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();
```

```
   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((11), 'University Rideshare Zenith
15', '2025-02-01', 600000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS8-4', 'Tech Nova 15', 'CubeSat', 8000, 7, (11), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((7), 'University Rideshare Nova 16',
'2025-02-01', 600000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS8-5', 'Academic Odin 16', 'CubeSat', 8000, 7, (7), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES ((8), 'University Rideshare Zenith
17', '2025-02-01', 600000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('CS8-6', 'Student Apex 17', 'CubeSat', 8000, 7, (8), 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 3, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Scientific 1', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Atlas Legacy-1', 1, 0, DATEADD(MONTH, 1, '2016-01-01'), DATEADD(MONTH, 1, '2016-01-01'), CASE WHEN 1 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 1 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P1', 'MicroSat Sirius 1', 'Satellite', 100000, 1, 1, CASE WHEN 1 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 1 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11001', CASE WHEN 1 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21001', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 2', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Chronos Legacy-2', 1, 0, DATEADD(MONTH, 2, '2016-01-01'), DATEADD(MONTH, 2, '2016-01-01'), CASE WHEN 2 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 2 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P2', 'MicroSat Apex 2', 'Satellite', 100000, 1, 1, CASE WHEN 2 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 2 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11002', CASE WHEN 2 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21002', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Legacy Gov Ops 3', '2015-01-01',
5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Luna Legacy-3', 1, 0, DATEADD(MONTH, 3, '2016-01-01'), DATEADD(MONTH,
3, '2016-01-01'), CASE WHEN 3 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 3 % 4 =
0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P3', 'MicroSat Orion 3', 'Satellite', 100000, 1, 1, CASE WHEN 3 %
4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 3 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11003', CASE WHEN 3 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21003', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Broadband Expansion 4', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Orion Legacy-4', 1, 0, DATEADD(MONTH, 4, '2016-01-01'), DATEADD(MONTH, 4, '2016-01-01'), CASE WHEN 4 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 4 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P4', 'MicroSat Apex 4', 'Satellite', 100000, 1, 1, CASE WHEN 4 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 4 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11004', CASE WHEN 4 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21004', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Broadband Expansion 5', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Ares Legacy-5', 1, 0, DATEADD(MONTH, 5, '2016-01-01'), DATEADD(MONTH, 5, '2016-01-01'), CASE WHEN 5 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 5 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P5', 'MicroSat Argus 5', 'Satellite', 100000, 1, 1, CASE WHEN 5 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 5 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11005', CASE WHEN 5 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21005', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 6', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Nyx Legacy-6', 1, 0, DATEADD(MONTH, 6, '2016-01-01'), DATEADD(MONTH, 6, '2016-01-01'), CASE WHEN 6 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 6 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P6', 'MicroSat Hermes 6', 'Satellite', 100000, 1, 1, CASE WHEN 6 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 6 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11006', CASE WHEN 6 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21006', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 7', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Titan Legacy-7', 1, 0, DATEADD(MONTH, 7, '2016-01-01'), DATEADD(MONTH, 7, '2016-01-01'), CASE WHEN 7 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 7 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P7', 'MicroSat Helios 7', 'Satellite', 100000, 1, 1, CASE WHEN 7 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 7 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11007', CASE WHEN 7 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21007', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Broadband Expansion 8', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Zeus Legacy-8', 1, 0, DATEADD(MONTH, 8, '2016-01-01'), DATEADD(MONTH, 8, '2016-01-01'), CASE WHEN 8 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 8 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P8', 'MicroSat Polaris 8', 'Satellite', 100000, 1, 1, CASE WHEN 8 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 8 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11008', CASE WHEN 8 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21008', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Gov Ops 9', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Vanguard Legacy-9', 1, 0, DATEADD(MONTH, 9, '2016-01-01'), DATEADD(MONTH, 9, '2016-01-01'), CASE WHEN 9 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 9 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P9', 'MicroSat Hermes 9', 'Satellite', 100000, 1, 1, CASE WHEN 9 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 9 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11009', CASE WHEN 9 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21009', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 10', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Echo Legacy-10', 1, 0, DATEADD(MONTH, 10, '2016-01-01'), DATEADD(MONTH, 10, '2016-01-01'), CASE WHEN 10 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 10 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P10', 'MicroSat Sol 10', 'Satellite', 100000, 1, 1, CASE WHEN 10 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 10 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11010', CASE WHEN 10 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21010', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 11', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Vega Legacy-11', 1, 0, DATEADD(MONTH, 11, '2016-01-01'), DATEADD(MONTH, 11, '2016-01-01'), CASE WHEN 11 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 11 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P11', 'MicroSat Atlas 11', 'Satellite', 100000, 1, 1, CASE WHEN 11 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 11 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11011', CASE WHEN 11 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21011', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 12',
'2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Hermes Legacy-12', 1, 0, DATEADD(MONTH, 12, '2016-01-01'),
DATEADD(MONTH, 12, '2016-01-01'), CASE WHEN 12 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1',
CASE WHEN 12 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P12', 'MicroSat Nyx 12', 'Satellite', 100000, 1, 1, CASE WHEN 12
% 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 12 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11012', CASE WHEN 12 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21012', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 13', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Polaris Legacy-13', 1, 0, DATEADD(MONTH, 13, '2016-01-01'), DATEADD(MONTH, 13, '2016-01-01'), CASE WHEN 13 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 13 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P13', 'MicroSat Sirius 13', 'Satellite', 100000, 1, 1, CASE WHEN 13 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 13 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11013', CASE WHEN 13 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21013', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 14', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Zeus Legacy-14', 1, 0, DATEADD(MONTH, 14, '2016-01-01'), DATEADD(MONTH, 14, '2016-01-01'), CASE WHEN 14 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 14 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P14', 'MicroSat Sol 14', 'Satellite', 100000, 1, 1, CASE WHEN 14 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 14 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11014', CASE WHEN 14 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21014', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 15',
'2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Luna Legacy-15', 1, 0, DATEADD(MONTH, 15, '2016-01-01'),
DATEADD(MONTH, 15, '2016-01-01'), CASE WHEN 15 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1',
CASE WHEN 15 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P15', 'MicroSat Ares 15', 'Satellite', 100000, 1, 1, CASE WHEN 15
% 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 15 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11015', CASE WHEN 15 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21015', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 16', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Zeus Legacy-16', 1, 0, DATEADD(MONTH, 16, '2016-01-01'), DATEADD(MONTH, 16, '2016-01-01'), CASE WHEN 16 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 16 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P16', 'MicroSat Hermes 16', 'Satellite', 100000, 1, 1, CASE WHEN 16 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 16 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11016', CASE WHEN 16 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21016', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Commercial Delivery 17', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Ares Legacy-17', 1, 0, DATEADD(MONTH, 17, '2016-01-01'), DATEADD(MONTH, 17, '2016-01-01'), CASE WHEN 17 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 17 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P17', 'MicroSat Atlas 17', 'Satellite', 100000, 1, 1, CASE WHEN 17 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 17 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11017', CASE WHEN 17 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21017', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Gov Ops 18', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Pioneer Legacy-18', 1, 0, DATEADD(MONTH, 18, '2016-01-01'), DATEADD(MONTH, 18, '2016-01-01'), CASE WHEN 18 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 18 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P18', 'MicroSat Ares 18', 'Satellite', 100000, 1, 1, CASE WHEN 18 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 18 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11018', CASE WHEN 18 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21018', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 19', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Titan Legacy-19', 1, 0, DATEADD(MONTH, 19, '2016-01-01'), DATEADD(MONTH, 19, '2016-01-01'), CASE WHEN 19 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 19 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P19', 'MicroSat Sirius 19', 'Satellite', 100000, 1, 1, CASE WHEN 19 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 19 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11019', CASE WHEN 19 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21019', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 20', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Echo Legacy-20', 1, 0, DATEADD(MONTH, 20, '2016-01-01'), DATEADD(MONTH, 20, '2016-01-01'), CASE WHEN 20 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 20 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P20', 'MicroSat Sirius 20', 'Satellite', 100000, 1, 1, CASE WHEN 20 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 20 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11020', CASE WHEN 20 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21020', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 21',
'2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Chronos Legacy-21', 1, 0, DATEADD(MONTH, 21, '2016-01-01'),
DATEADD(MONTH, 21, '2016-01-01'), CASE WHEN 21 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1',
CASE WHEN 21 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P21', 'MicroSat Nyx 21', 'Satellite', 100000, 1, 1, CASE WHEN 21
% 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 21 % 4 = 0 THEN 3
ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 1', 'EXC11-11021', CASE WHEN 21 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 1 Stage 2', 'EXC11-21021', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Scientific 22', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Nova Legacy-22', 1, 0, DATEADD(MONTH, 22, '2016-01-01'), DATEADD(MONTH, 22, '2016-01-01'), CASE WHEN 22 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 22 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P22', 'MicroSat Orion 22', 'Satellite', 100000, 1, 1, CASE WHEN 22 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 22 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11022', CASE WHEN 22 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21022', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Legacy Global Network 23', '2015-01-01', 5000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Pioneer Legacy-23', 1, 0, DATEADD(MONTH, 23, '2016-01-01'), DATEADD(MONTH, 23, '2016-01-01'), CASE WHEN 23 % 4 = 0 THEN 3 ELSE 2 END, 'EXC1', CASE WHEN 23 % 4 = 0 THEN 'Failure' ELSE 'Success' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('EXC1-P23', 'MicroSat Nyx 23', 'Satellite', 100000, 1, 1, CASE WHEN 23 % 4 = 0 THEN 28 ELSE 37 END);

   SET @PayloadID = SCOPE_IDENTITY();
```

```
   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 23 % 4 = 0 THEN 3 ELSE 2 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 1', 'EXC11-11023', CASE WHEN 23 % 4 = 0 THEN 28 ELSE 30 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior 1 Stage 2', 'EXC11-21023', 30);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 1', '2023-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Nova Prototype Test 1', 5, 0, DATEADD(MONTH, 1, '2024-01-01'), DATEADD(MONTH, 1, '2024-01-01'), CASE WHEN 1 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova Test', CASE WHEN 1 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('NOVA-P1', 'Nova Mass Sim 1', 'Test Mass', 50000000, 1, 1, CASE WHEN 1 IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 1 IN (3, 5, 7) THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior Nova Stage 1', 'EXCN1-10008', CASE WHEN 1 IN (3, 5, 7) THEN 1 ELSE 28 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior Nova Stage 2', 'EXCN1-20008', CASE WHEN 1 IN (3, 5, 7) THEN 1 ELSE 28 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');
```

```
   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 2', '2023-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Nova Prototype Test 2', 5, 0, DATEADD(MONTH, 2, '2024-01-01'), DATEADD(MONTH, 2, '2024-01-01'), CASE WHEN 2 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova Test', CASE WHEN 2 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('NOVA-P2', 'Nova Mass Sim 2', 'Test Mass', 50000000, 1, 1, CASE WHEN 2 IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id, deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 2 IN (3, 5, 7) THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES (@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior Nova Stage 1', 'EXCN1-10009', CASE WHEN 2 IN (3, 5, 7) THEN 1 ELSE 28 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES ('Excelsior Nova Stage 2', 'EXCN1-20009', CASE WHEN 2 IN (3, 5, 7) THEN 1 ELSE 28 END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number, stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date, contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 3', '2023-01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission, launch_datetime_net, launch_datetime_scheduled, mission_status_code, mission_description, post_mission_summary)

   VALUES ('Nova Prototype Test 3', 5, 0, DATEADD(MONTH, 3, '2024-01-01'), DATEADD(MONTH, 3, '2024-01-01'), CASE WHEN 3 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova Test', CASE WHEN 3 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type, payload_mass_g, manufacturer_id, customer_id, payload_status_code)
```

```
   VALUES ('NOVA-P3', 'Nova Mass Sim 3', 'Test Mass', 50000000, 1, 1, CASE WHEN 3
IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 3 IN (3, 5, 7)
THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 1', 'EXCN1-10003', CASE WHEN 3 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 2', 'EXCN1-20003', CASE WHEN 3 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 4', '2023-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Nova Prototype Test 4', 5, 0, DATEADD(MONTH, 4, '2024-01-01'),
DATEADD(MONTH, 4, '2024-01-01'), CASE WHEN 4 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova
Test', CASE WHEN 4 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('NOVA-P4', 'Nova Mass Sim 4', 'Test Mass', 50000000, 1, 1, CASE WHEN 4
IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 4 IN (3, 5, 7)
THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 1', 'EXCN1-10004', CASE WHEN 4 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');
```

```
   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 2', 'EXCN1-20004', CASE WHEN 4 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 5', '2023-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Nova Prototype Test 5', 5, 0, DATEADD(MONTH, 5, '2024-01-01'),
DATEADD(MONTH, 5, '2024-01-01'), CASE WHEN 5 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova
Test', CASE WHEN 5 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('NOVA-P5', 'Nova Mass Sim 5', 'Test Mass', 50000000, 1, 1, CASE WHEN 5
IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 5 IN (3, 5, 7)
THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 1', 'EXCN1-10005', CASE WHEN 5 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 2', 'EXCN1-20005', CASE WHEN 5 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 6', '2023-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)
```

```
   VALUES ('Nova Prototype Test 6', 5, 0, DATEADD(MONTH, 6, '2024-01-01'),
DATEADD(MONTH, 6, '2024-01-01'), CASE WHEN 6 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova
Test', CASE WHEN 6 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('NOVA-P6', 'Nova Mass Sim 6', 'Test Mass', 50000000, 1, 1, CASE WHEN 6
IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 6 IN (3, 5, 7)
THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 1', 'EXCN1-10006', CASE WHEN 6 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 2', 'EXCN1-20006', CASE WHEN 6 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (1, 'Avantvia Dev Contract 7', '2023-
01-01', 50000000.00, '...');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Nova Prototype Test 7', 5, 0, DATEADD(MONTH, 7, '2024-01-01'),
DATEADD(MONTH, 7, '2024-01-01'), CASE WHEN 7 IN (3, 5, 7) THEN 2 ELSE 3 END, 'Nova
Test', CASE WHEN 7 IN (3, 5, 7) THEN 'Success' ELSE 'Failure' END);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('NOVA-P7', 'Nova Mass Sim 7', 'Test Mass', 50000000, 1, 1, CASE WHEN 7
IN (3, 5, 7) THEN 37 ELSE 28 END);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 1, CASE WHEN 7 IN (3, 5, 7)
THEN 2 ELSE 3 END);

   SET @ManifestID = SCOPE_IDENTITY();
```

```
   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 1', 'EXCN1-10007', CASE WHEN 7 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Stage 1');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior Nova Stage 2', 'EXCN1-20007', CASE WHEN 7 IN (3, 5, 7) THEN 1 ELSE 28
END);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   INSERT INTO contracts (contractor_id, contract_title, contract_signed_date,
contract_value_eur, contract_details) VALUES (2, 'NASA Artemis Gateway Logistics',
GETDATE(), 4000000000.00, 'Gateway');

   SET @ContractID = SCOPE_IDENTITY();

   INSERT INTO missions (mission_name, launch_pad_id, is_crewed_mission,
launch_datetime_net,        launch_datetime_scheduled,        mission_status_code,
mission_description, post_mission_summary)

   VALUES ('Artemis Gateway-1', 1, 0, DATEADD(YEAR, 1, GETDATE()), DATEADD(YEAR,
1, GETDATE()), 8, 'Gateway Module', NULL);

   SET @MissionID = SCOPE_IDENTITY();

   INSERT INTO payloads (payload_serial, payload_name, payload_type,
payload_mass_g, manufacturer_id, customer_id, payload_status_code)

   VALUES ('GW-1', 'Gateway Hab', 'Lander', 15000000, 1, 2, 1);

   SET @PayloadID = SCOPE_IDENTITY();

   INSERT INTO payload_mission_manifests (mission_id, payload_id, target_orbit_id,
deploy_status_code) VALUES (@MissionID, @PayloadID, 6, 5);

   SET @ManifestID = SCOPE_IDENTITY();

   INSERT INTO contract_manifest_links (manifest_id, contract_id) VALUES
(@ManifestID, @ContractID);

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 1', 'EXC23-11010', 1);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 1, 'Heavy Core');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 4, 1, 'Heavy Port');

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, 5, 1, 'Heavy Starboard');

   INSERT INTO rocket_stages (stage_type, stage_serial, stage_status_code) VALUES
('Excelsior 2 Stage 2', 'EXC23-29001', 1);

   INSERT INTO mission_rocket_stages (mission_id, stage_id, stage_number,
stage_role) VALUES (@MissionID, SCOPE_IDENTITY(), 2, 'Stage 2');


   COMMIT TRANSACTION;
```

```
PRINT 'Data successfully inserted.';
END TRY
BEGIN CATCH
    ROLLBACK TRANSACTION;
    PRINT ERROR_MESSAGE();
END CATCH

EXEC sp_MSForEachTable 'ALTER TABLE ? WITH NOCHECK CHECK CONSTRAINT ALL';
GO
```